PENGOLAHAN CITRA & VISI KOMPUTER

FFT & Filter domain frekuensi

(sumber: R.C. Gonzalez & R.E. Woods, Digital Image Processing, 4th edition)

Handayani Tjandrasa

Transformasi Fourier 1D



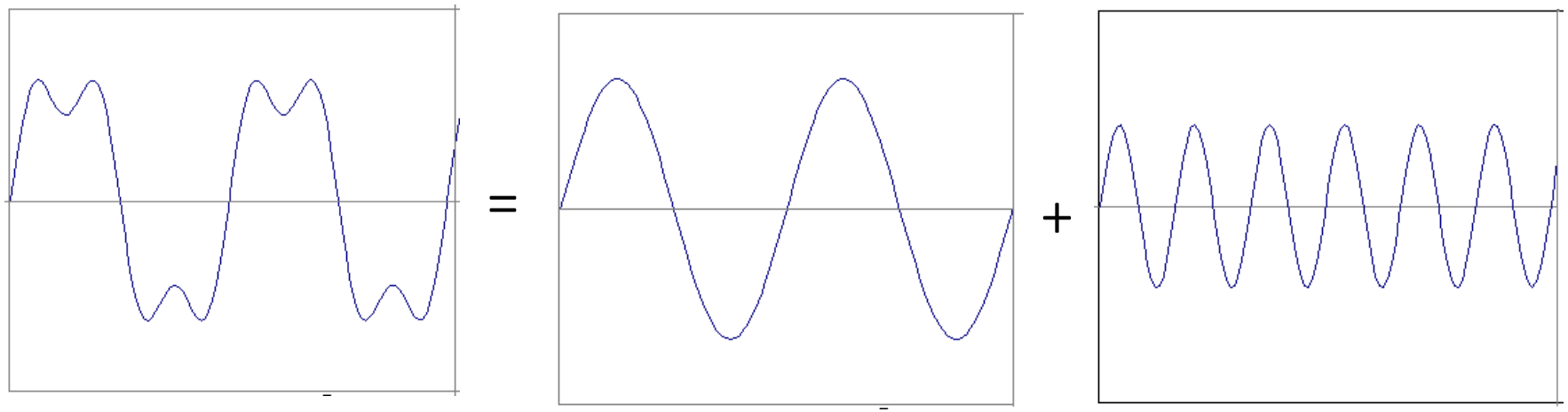
$$F(\omega) = R(\omega) + iI(\omega)$$

$$A = \pm\sqrt{R(\omega)^2 + I(\omega)^2} \qquad \phi = \tan^{-1}\frac{I(\omega)}{R(\omega)}$$

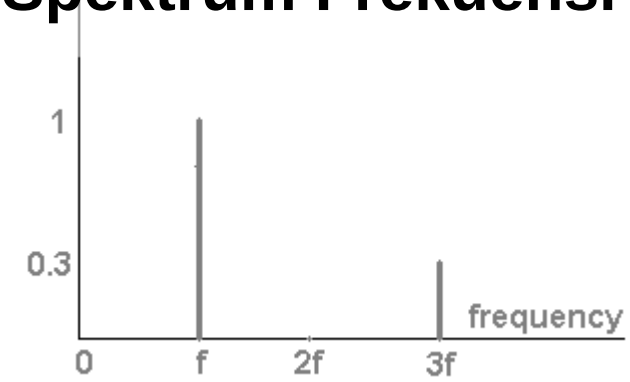


Sinyal (1D) dalam domain waktu dan domain frekuensi

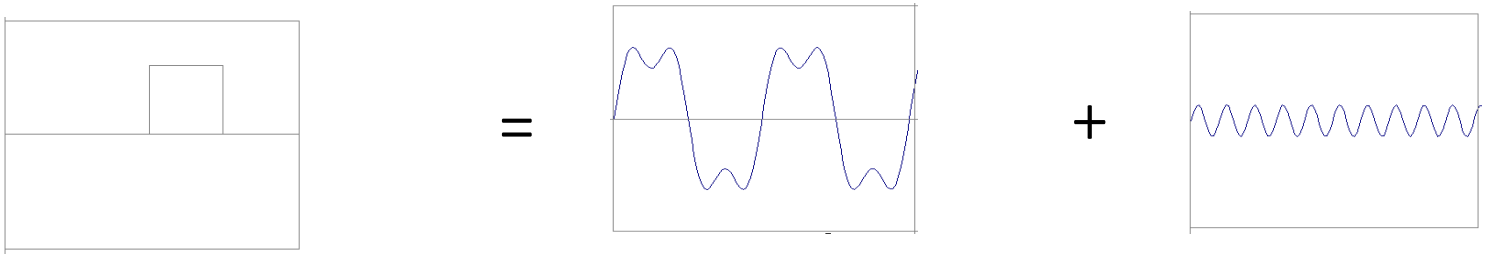
$$g(t) = \sin(2\pi f\, t) + (1/3)\sin(2\pi(3f)\, t)$$

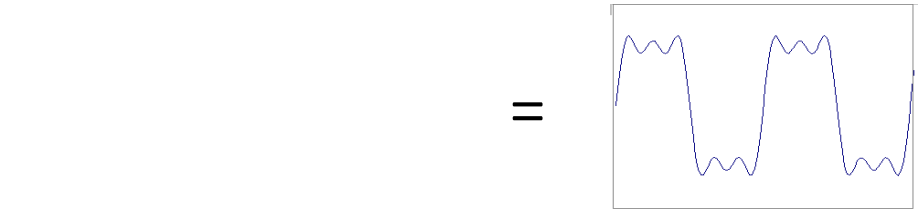


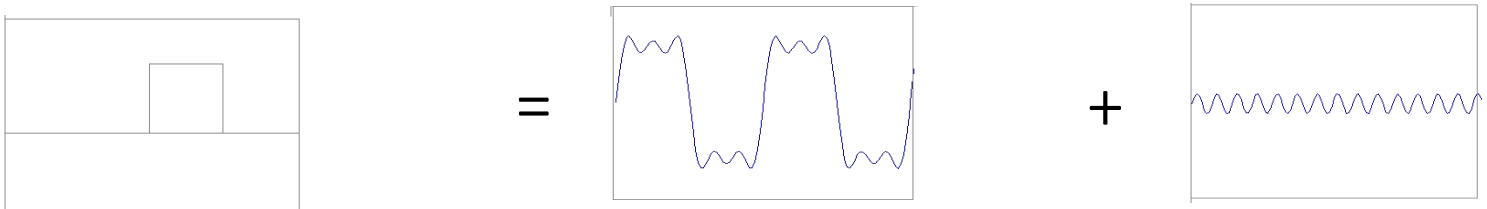
Spektrum Frekuensi

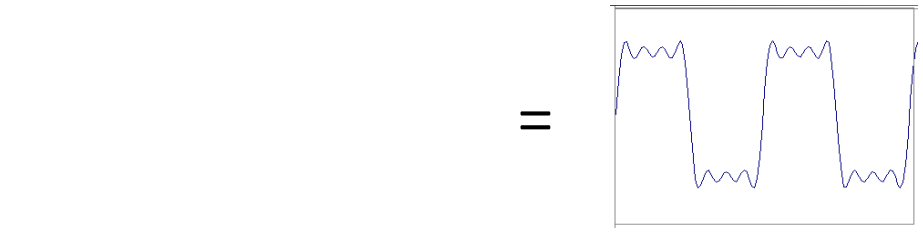


Superposisi

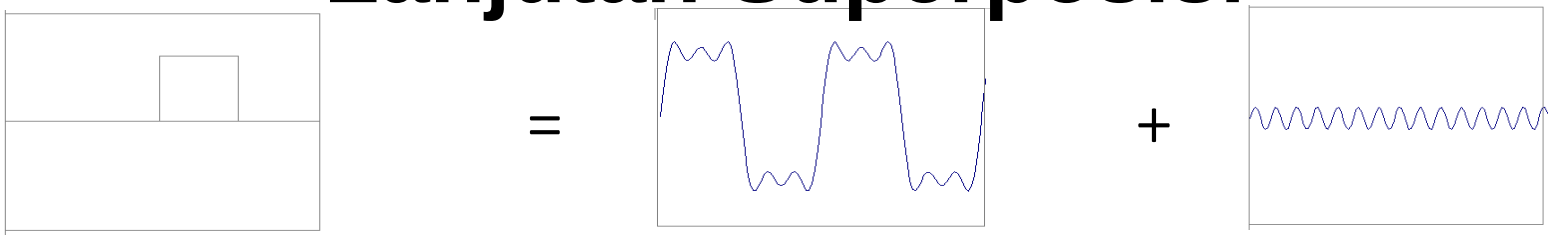


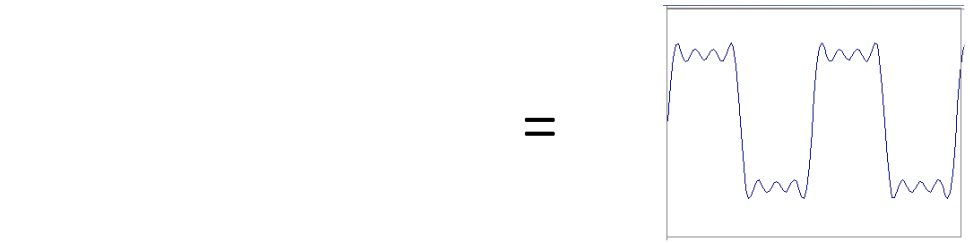


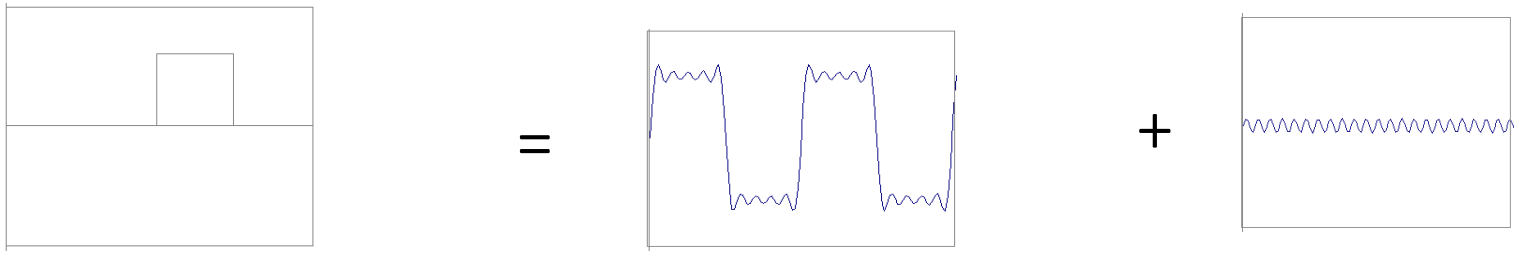


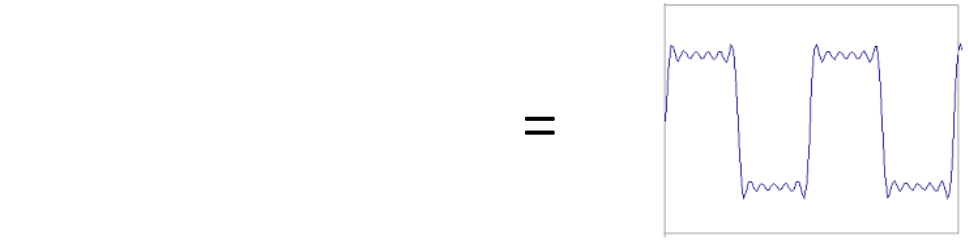


Lanjutan Superposisi

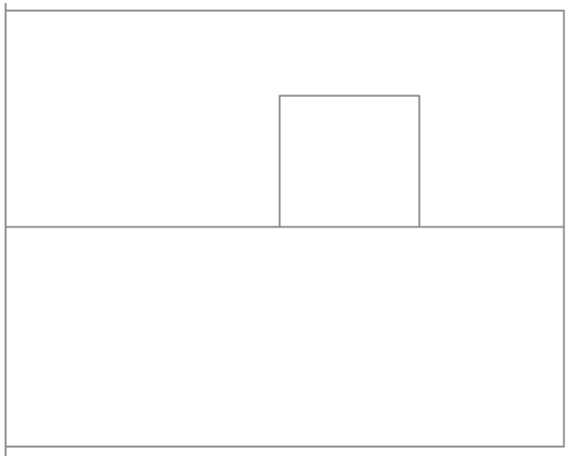






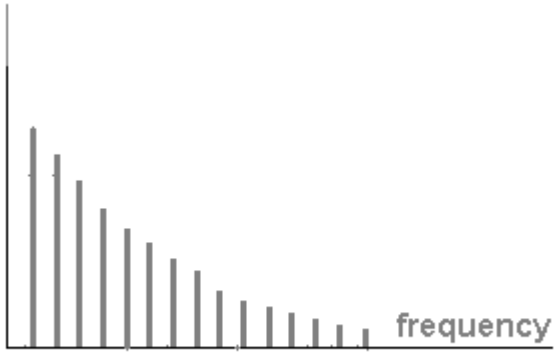


Spektrum frekuensi

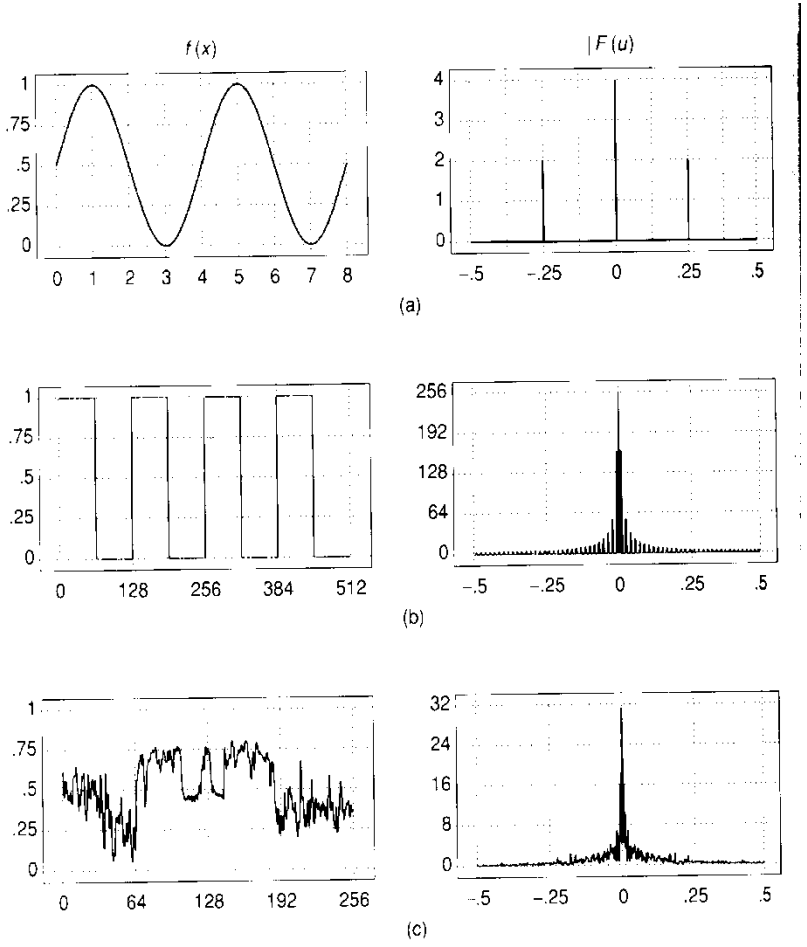


=

$$A\sum_{k=1}^{\infty}\frac{1}{k}\sin(2\pi kt)$$



Spektrum frekuensi



Transformasi Fourier kontinu 1D & Inverse

$$\text{Fourier Transform}: \quad F(\omega) = \int_{-\infty}^{+\infty} f(x) e^{-i\omega x} dx$$

$$\text{Inverse Fourier Transform}: \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) e^{i\omega x} d\omega$$

1D Discrete Fourier Transform

$$f(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos(\tfrac{2\pi kt}{T}) + \sum_{k=1}^{\infty} b_k \sin(\tfrac{2\pi kt}{T})$$

$$f(t) \approx a_0 + \sum_{k=1}^{m} a_k \cos(\tfrac{2\pi kt}{T}) + \sum_{k=1}^{m} b_k \sin(\tfrac{2\pi kt}{T})$$

Dalam bentuk bilangan kompleks:

$$\cos(k\theta) = \frac{e^{ik\theta} + e^{-ik\theta}}{2}, \quad \sin(k\theta) = \frac{e^{ik\theta} - e^{-ik\theta}}{2i}$$

$$f(t) \approx \sum_{k=-m}^{m} c_k e^{\frac{2\pi ikt}{T}} = \sum_{k=0}^{2m} c_k e^{\frac{2\pi ikt}{T}}$$

1D DFT

$$F_k = \frac{1}{N}\sum_{n=0}^{N-1} f_n W_N^{-nk} \qquad k = 0, 1, \ldots, N-1.$$

$$f_n = \sum_{k=0}^{N-1} F_k W_N^{nk} \qquad n = 0, 1, \ldots, N-1.$$

Dimana: $W_N = e^{2\pi i/N}$

Kompleksitas = $O(N^2)$

1D FFT (Fast Fourier Transform)

$$\mathrm{DFT}(f,N) \quad \leftrightarrow \quad F_k = \frac{1}{N}\sum_{n=0}^{N-1} f_n W_N^{-nk}$$

$$\frac{1}{N}\sum_{n=0}^{N/2-1} g_n W_{N/2}^{-nk} \qquad \frac{1}{N}\sum_{n=0}^{N/2-1} h_n W_{N/2}^{-nk}$$

$$\updownarrow \qquad\qquad \updownarrow$$

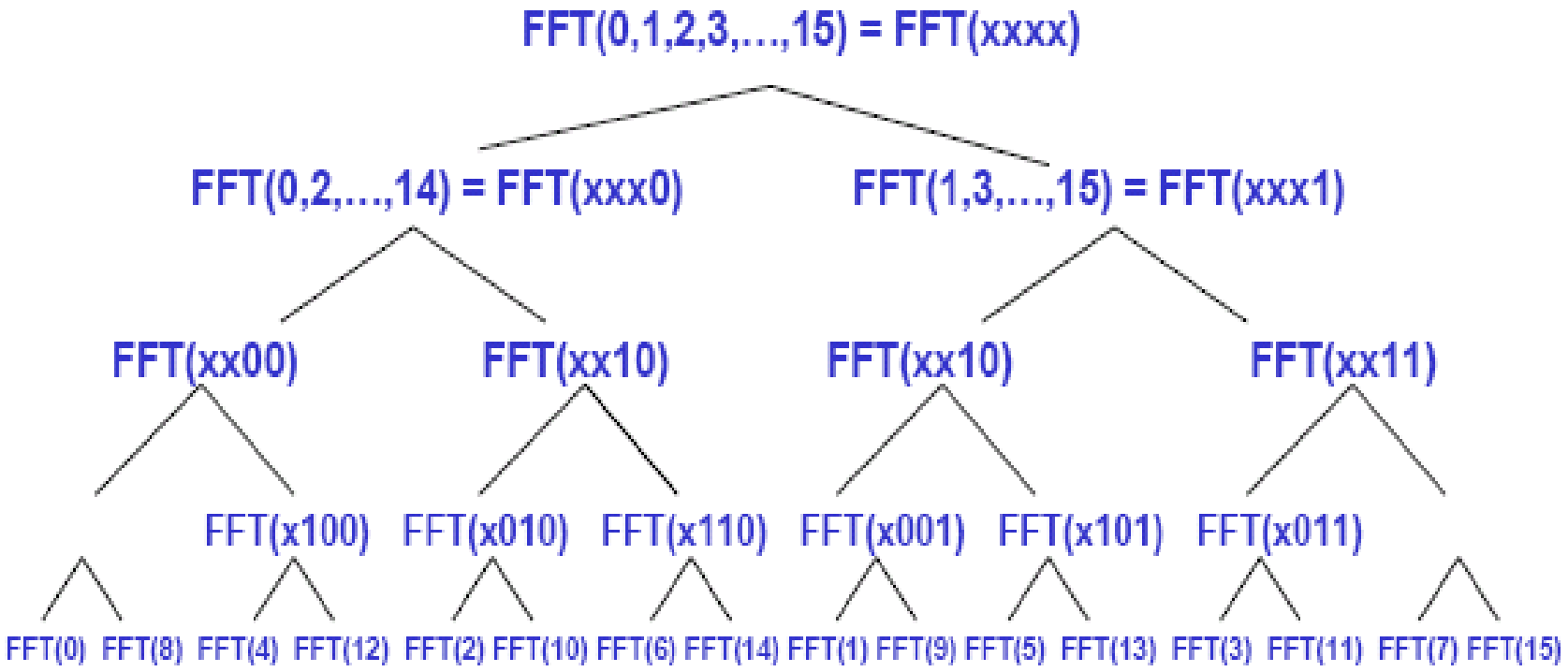
$$\mathrm{DFT}(g,N/2) \qquad \mathrm{DFT}(h,N/2)$$

$$F_{2k} = \frac{1}{N}\sum_{n=0}^{N/2-1} \underbrace{(f_n + f_{n+N/2})}_{g_n} W_{N/2}^{-nk} = \mathrm{DFT}(g,N/2)$$

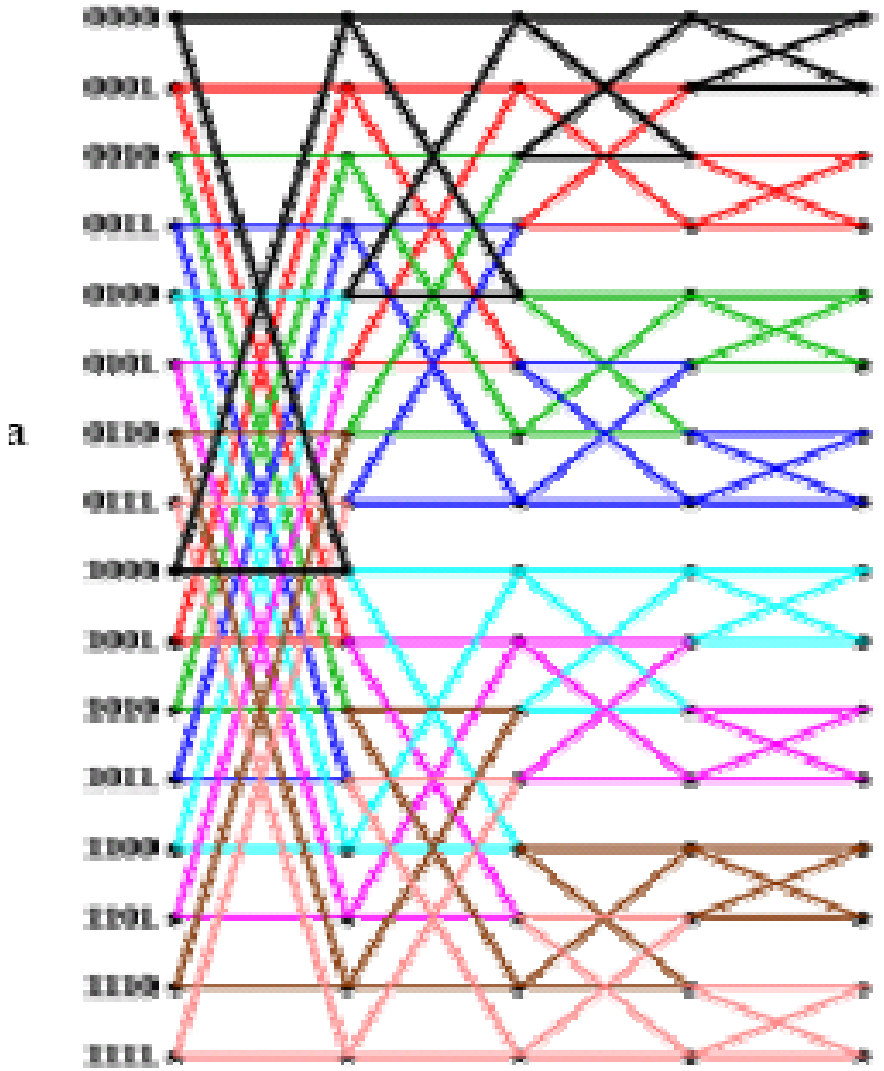
$$F_{2k+1} = \frac{1}{N}\sum_{n=0}^{N/2-1} \underbrace{[(f_n - f_{n+N/2})W^{-n}]}_{h_n} W_{N/2}^{-nk} = \mathrm{DFT}(h,N/2)$$

Komputasi DFT dari N koefisien = komputasi 2 DFT dari N/2 koefisien. FFT menghitung secara rekursif dengan kompleksitas: $\mathrm{O}(N \log_2 N)$

Diagram 1D FFT



Butterfly FFT Algorithm



1D DFT & 2D DFT

Bila 1D DFT dan inverse (1D IDFT) didefinisikan dengan konstanta sebagai berikut:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \qquad u = 0, 1, 2, \ldots, M-1$$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \qquad x = 0, 1, 2, \ldots, M-1$$

Maka untuk 2D DFT dan inverse (2D IDFT) dapat dinyatakan dengan :

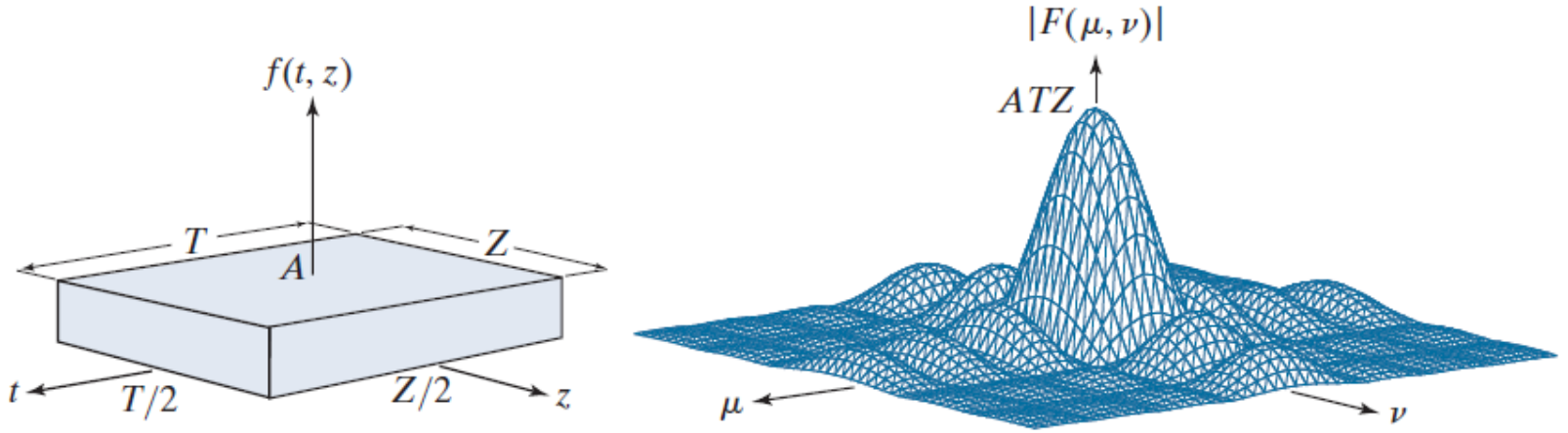
$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi(ux/M + vy/N)}$$

CITRA (2D) & spektrum frekuensi 2D

a b

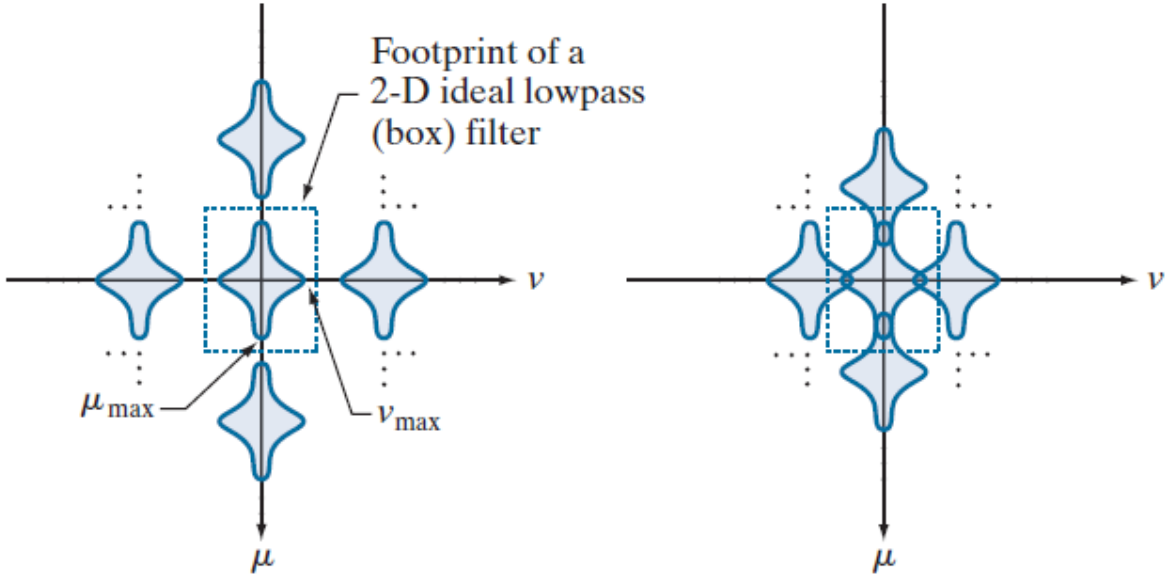
FIGURE 4.14
(a) A 2-D function and (b) a section of its spectrum. The box is longer along the t-axis, so the spectrum is more contracted along the μ-axis.



Over-sampled & under-sampled

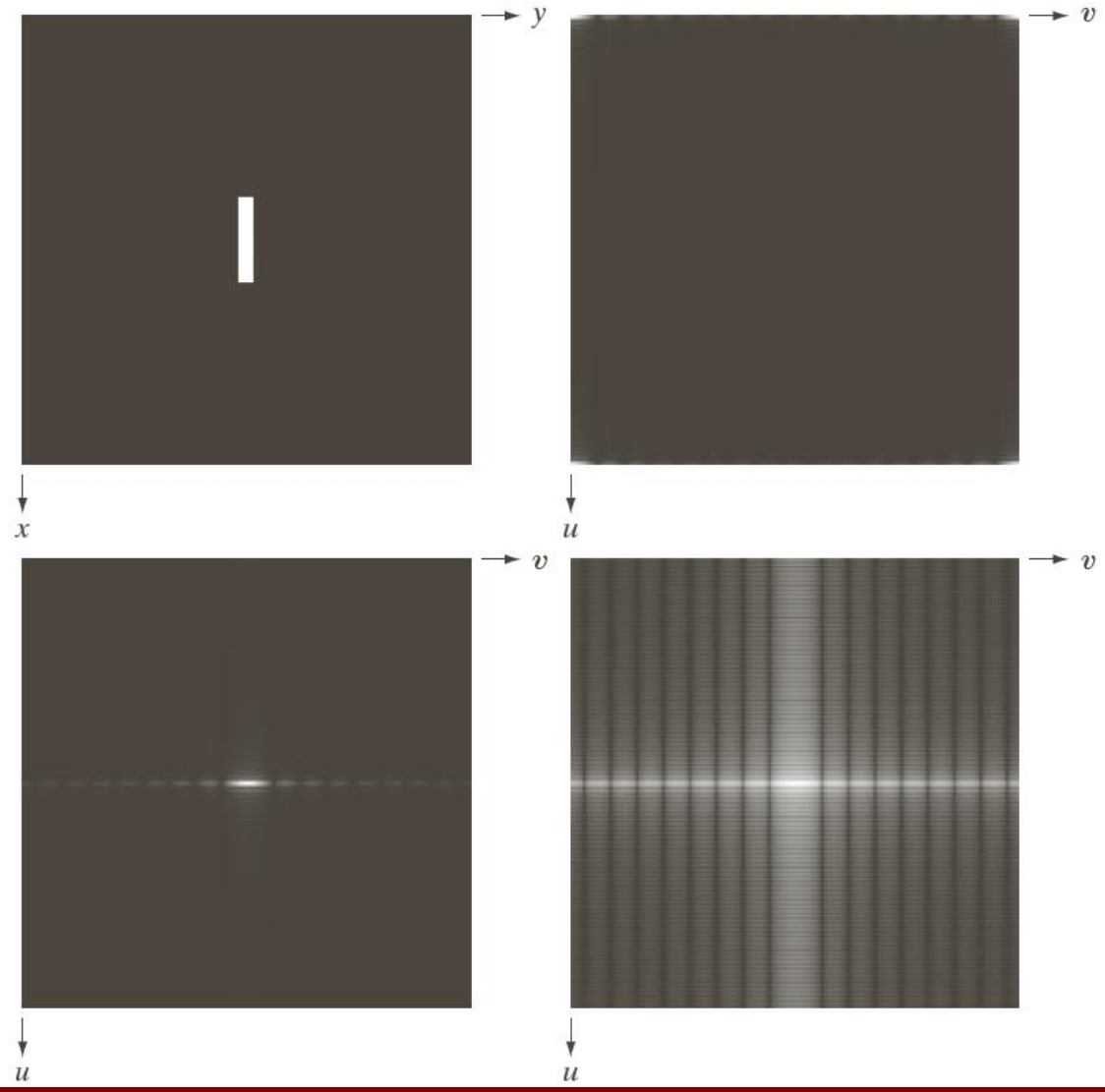
a b

FIGURE 4.16
Two-dimensional Fourier transforms of (a) an over-sampled, and (b) an under-sampled, band-limited function.



Langkah-langkah untuk menampilkan 2D FFT

- **Gunakan fftshift untuk menggeser ke 4 ujung kuadran ke tengah.**
- **Tingkatkan detail dengan fungsi log.**



a b
c d

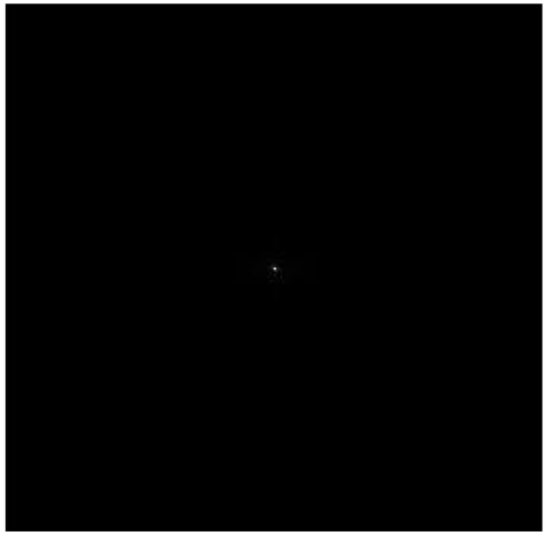
FIGURE 4.24
(a) Image. (b) Spectrum showing bright spots in the four corners. (c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.

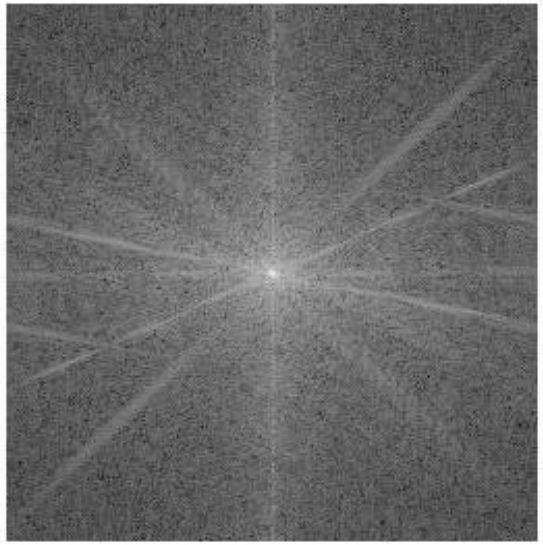
Program Matlab

```
clear all, close all;
% Read test image
img=imread('cameraman.tif');
figure('Name', 'input image'); imshow(img);
% Compute Fourier Transform
Fimg= fft2(img);
% Shift the spectrum to the center
Fs=fftshift(abs(Fimg));
figure('Name', 'center shifted fft'); imshow(Fs,[]);
% Compute log of spectrum
Fslog=log(Fs+1);
figure('Name','log fft'); imshow(Fslog,[]);
% The inverse of Fourier Transform
im=uint8(ifft2(Fimg));
figure('Name','im'); imshow(im);
```

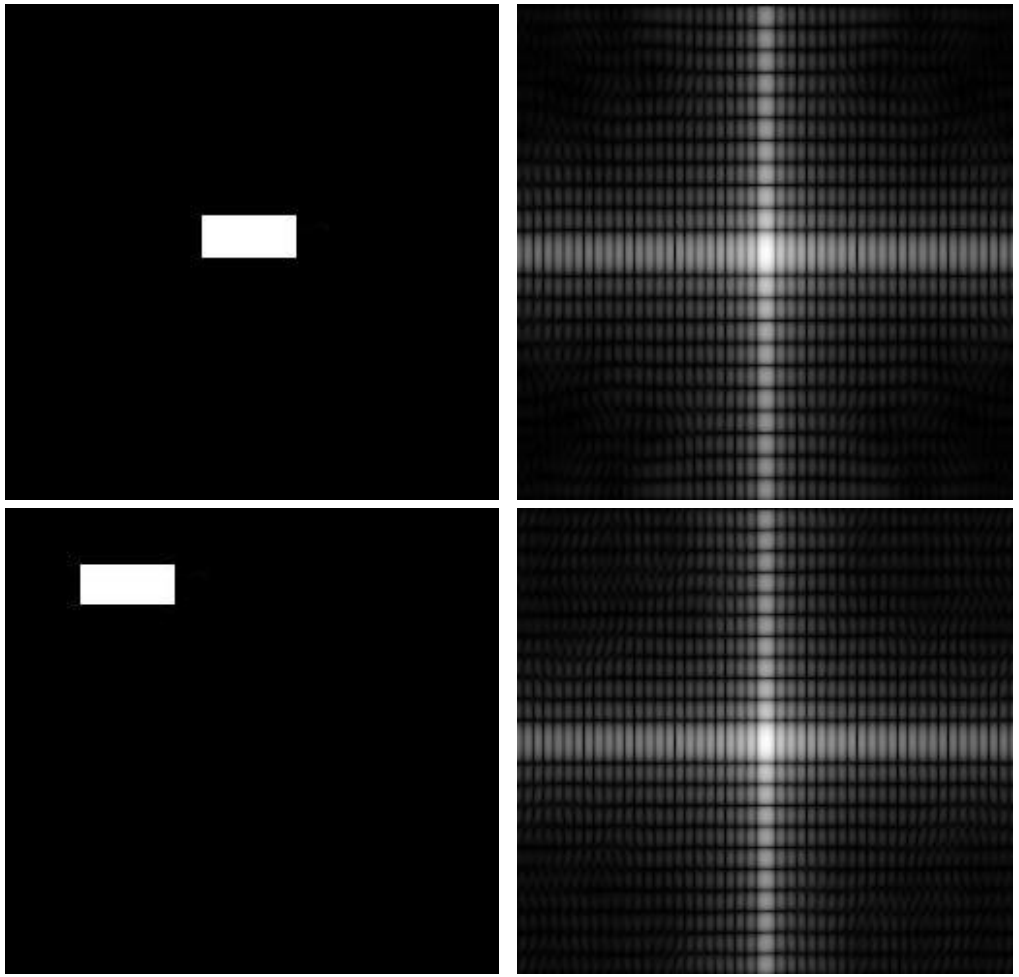
Hasil



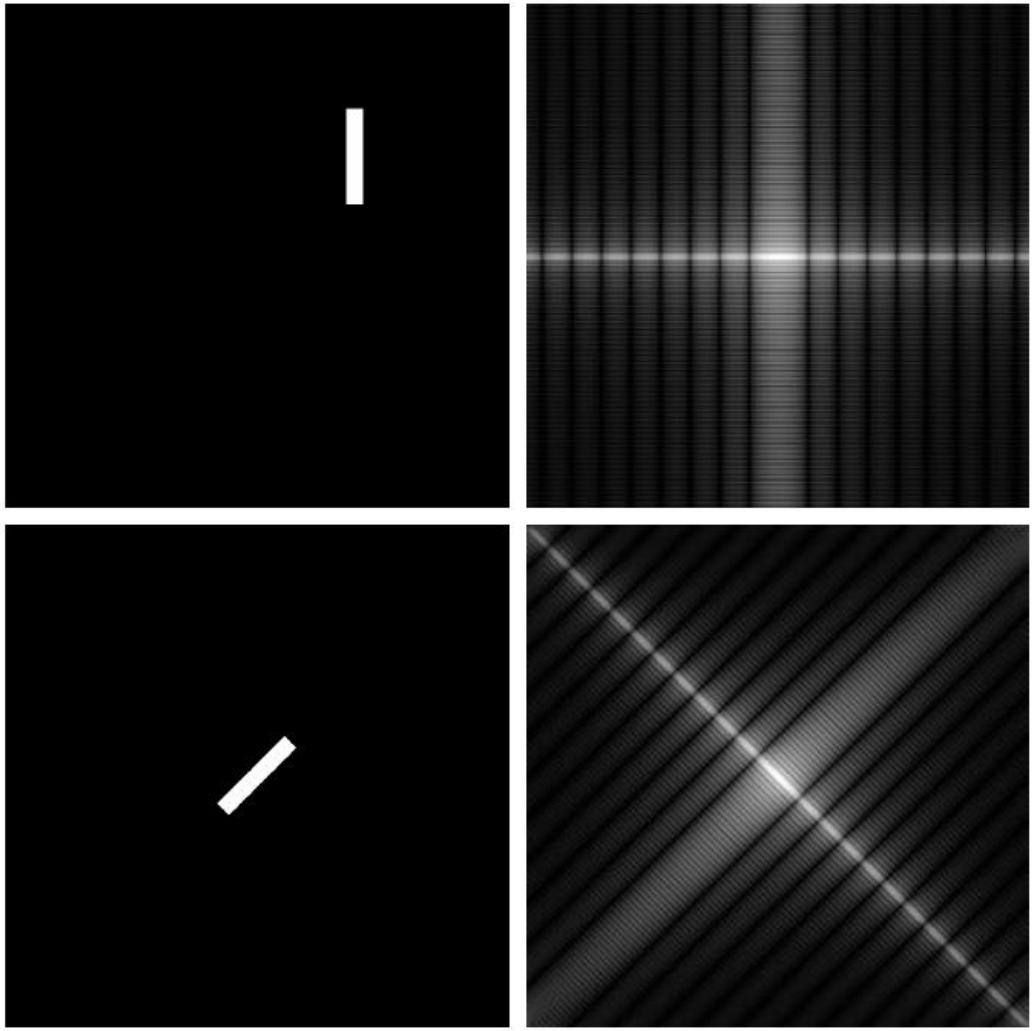




CITRA & 2D FFT



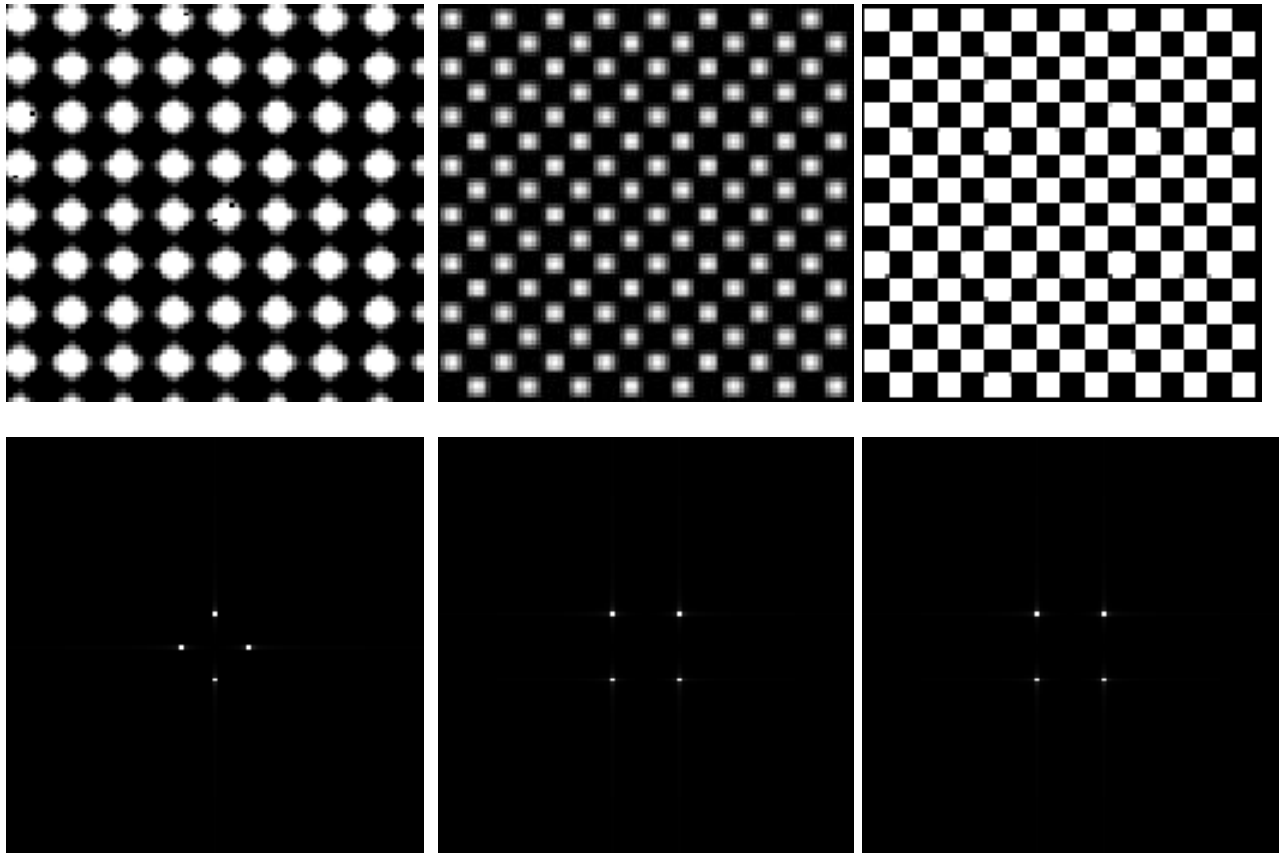
CITRA & 2D FFT



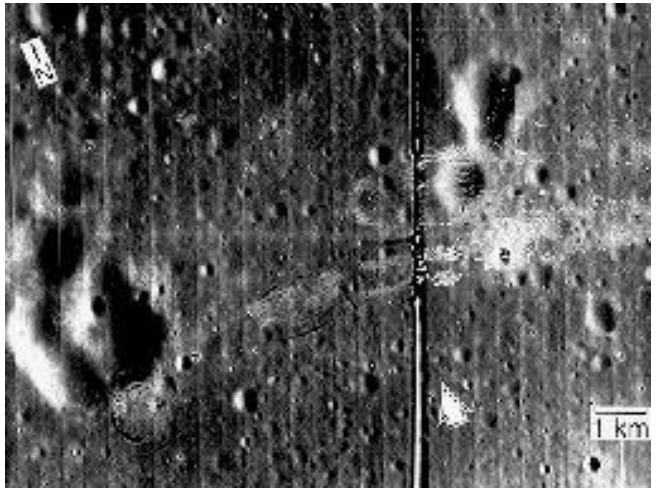
a b
c d

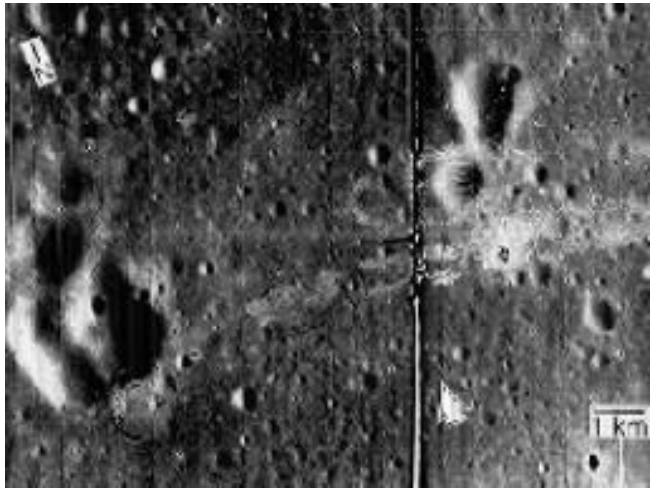
FIGURE 4.25
(a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum. (c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).

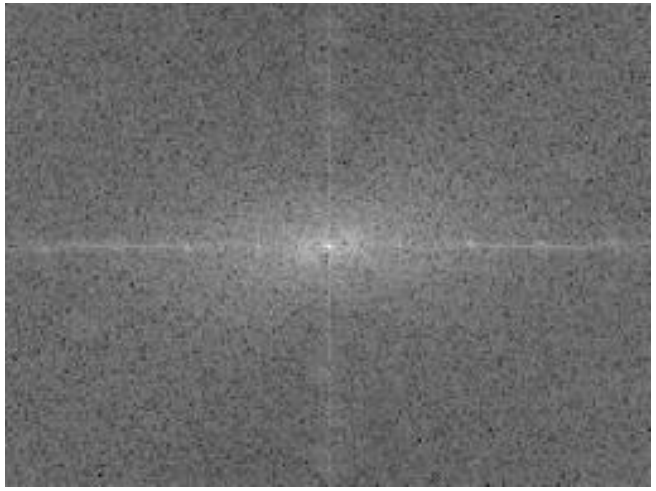
CITRA & 2D FFT

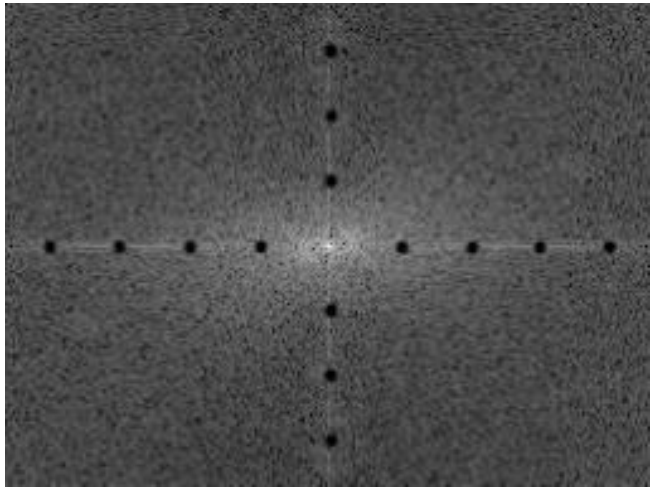


CITRA & 2D FFT



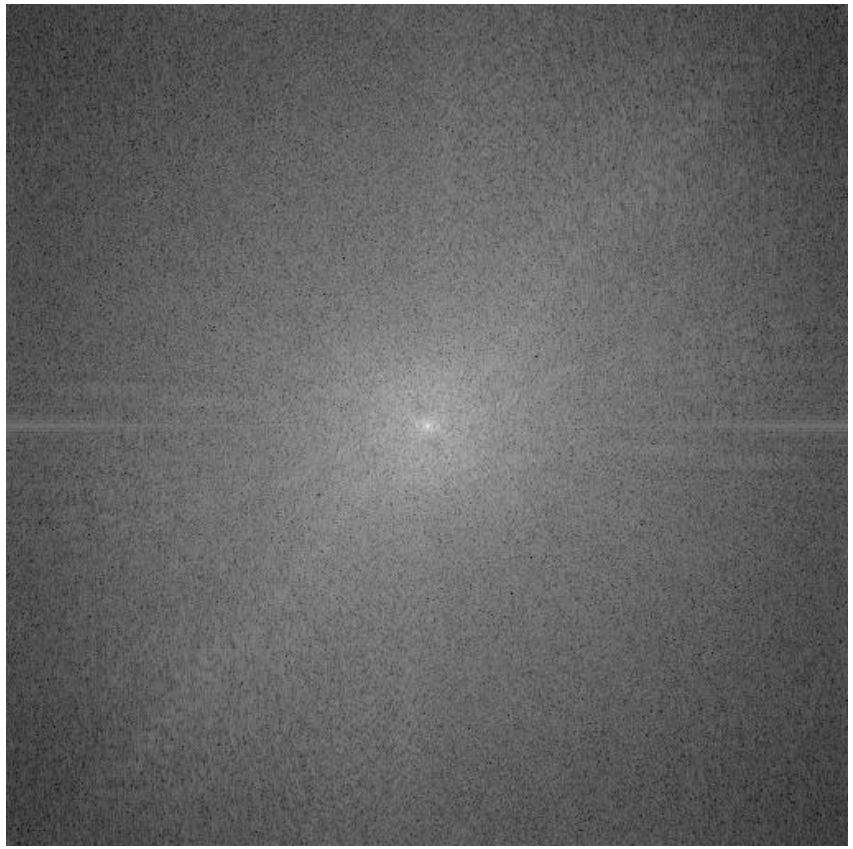






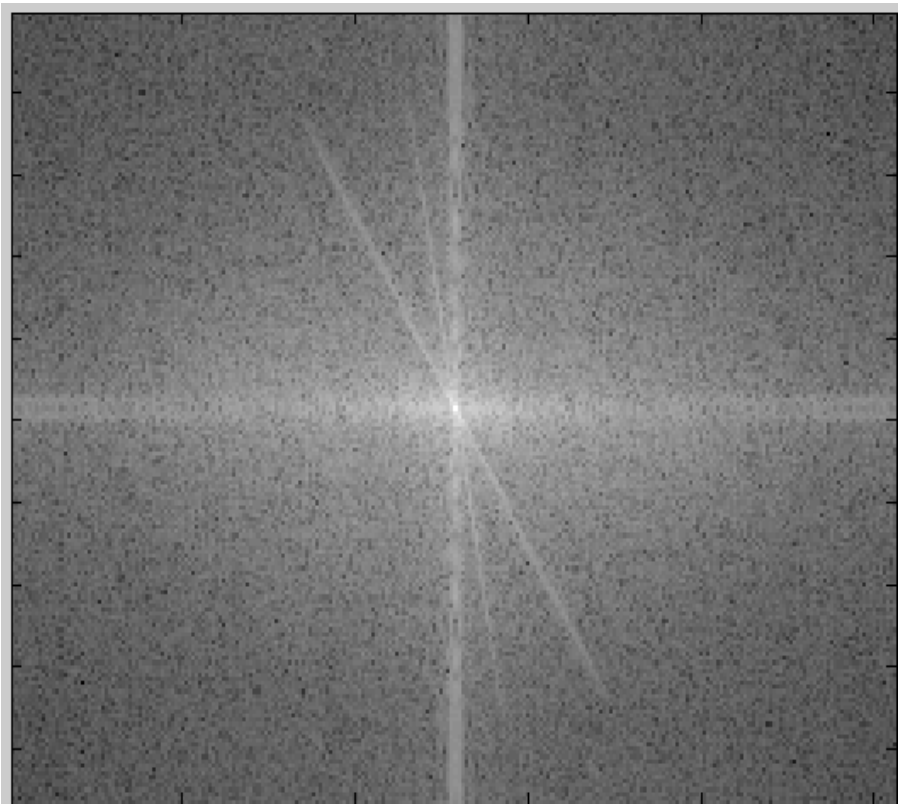
CITRA & 2D FFT



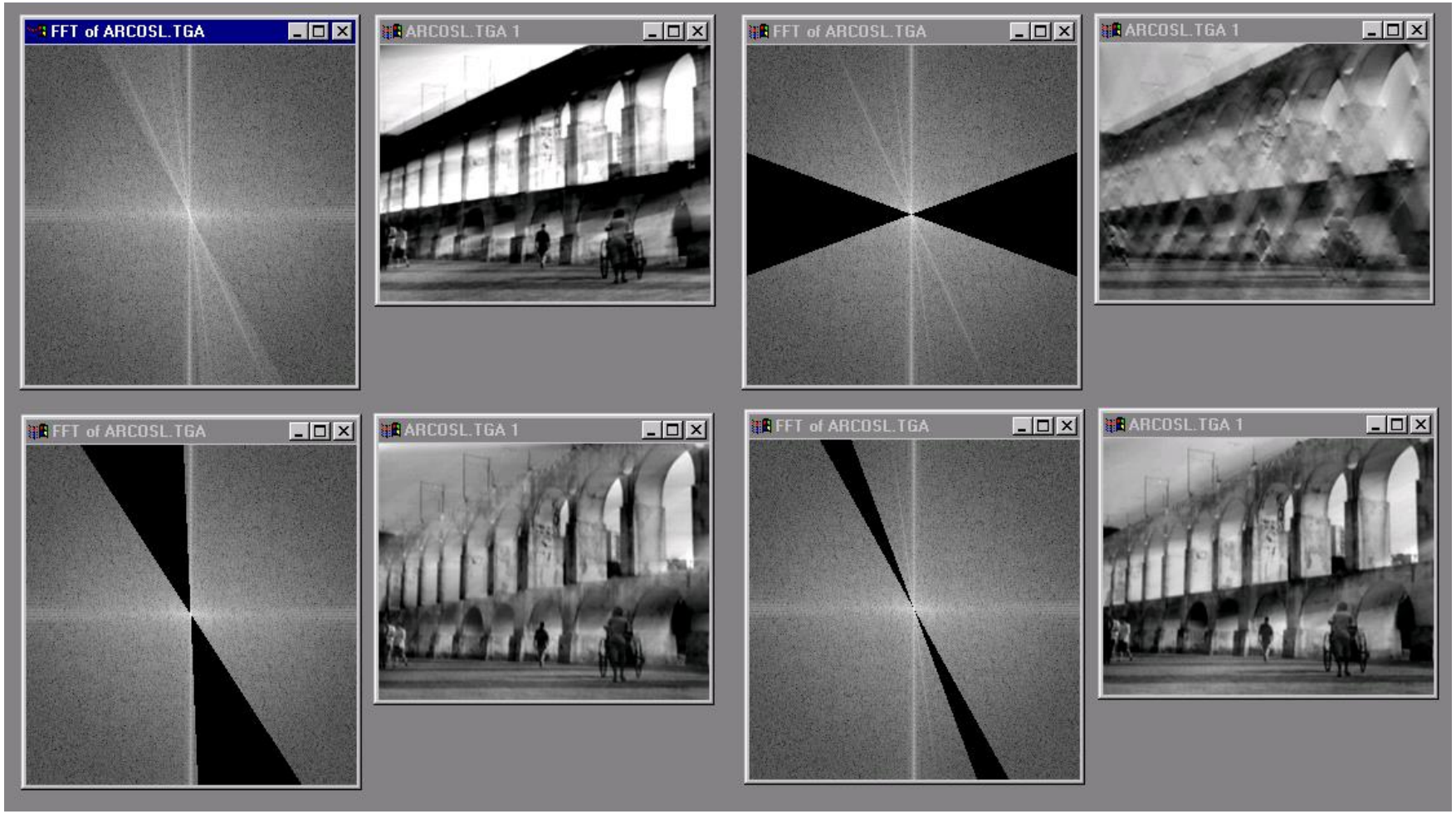


CITRA & 2D FFT

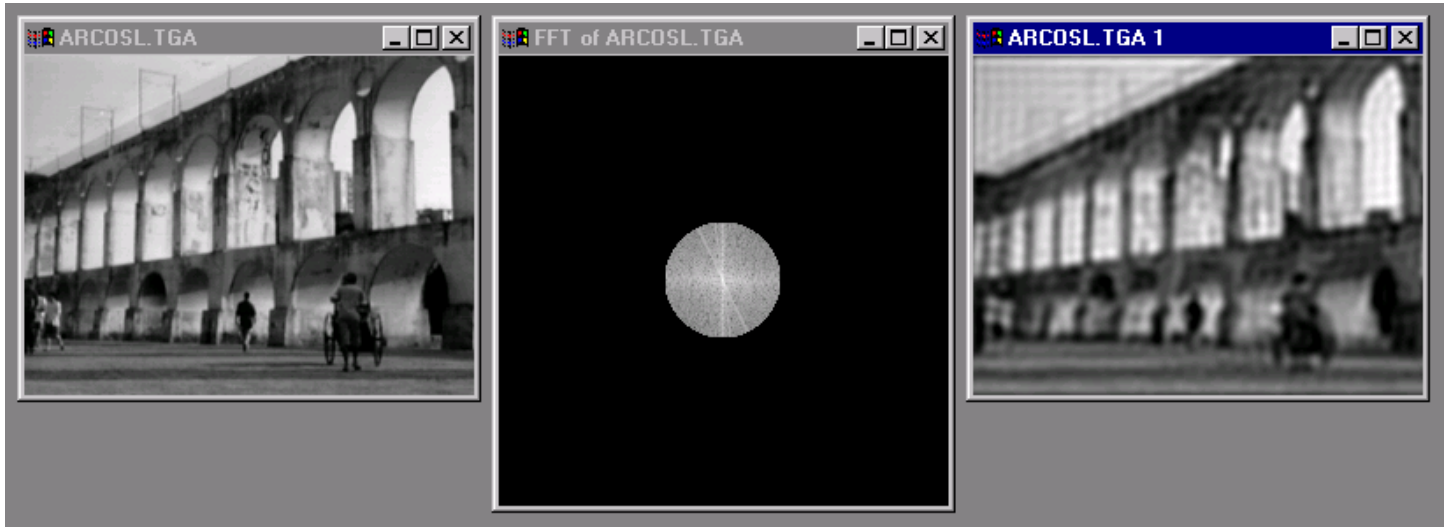




Spektrum diubah, sebelum direkontruksi



Informasi terkonsentrasi pada frekuensi rendah



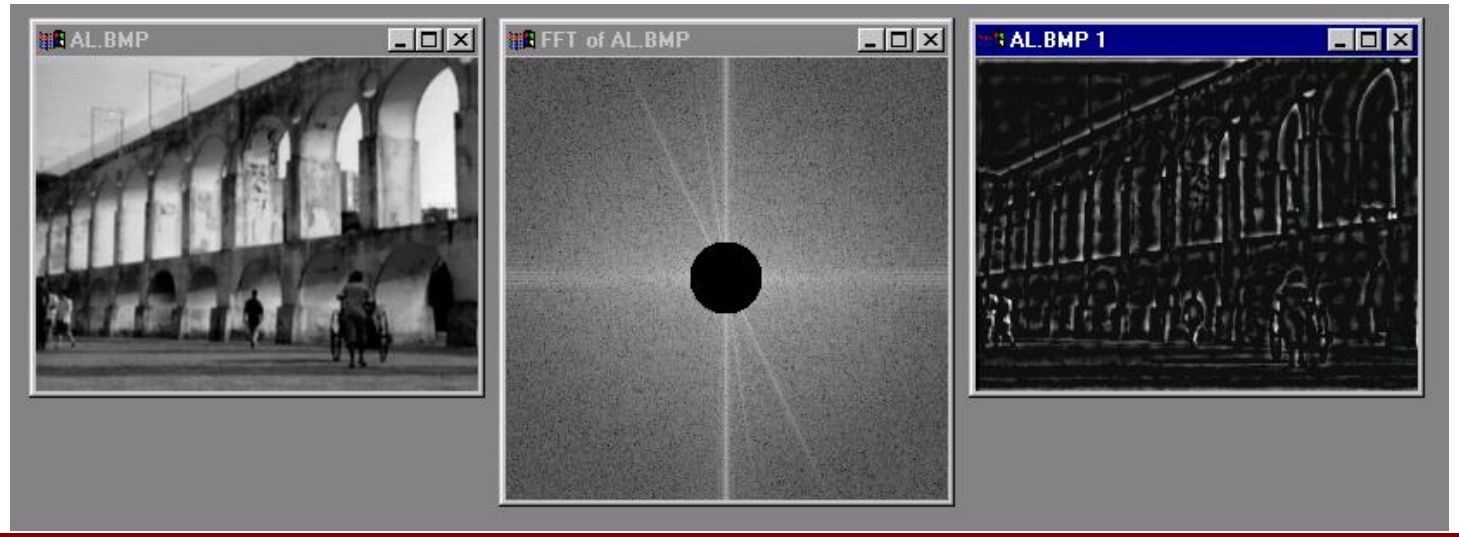


Diagram pemrosesan dalam domain frekuensi

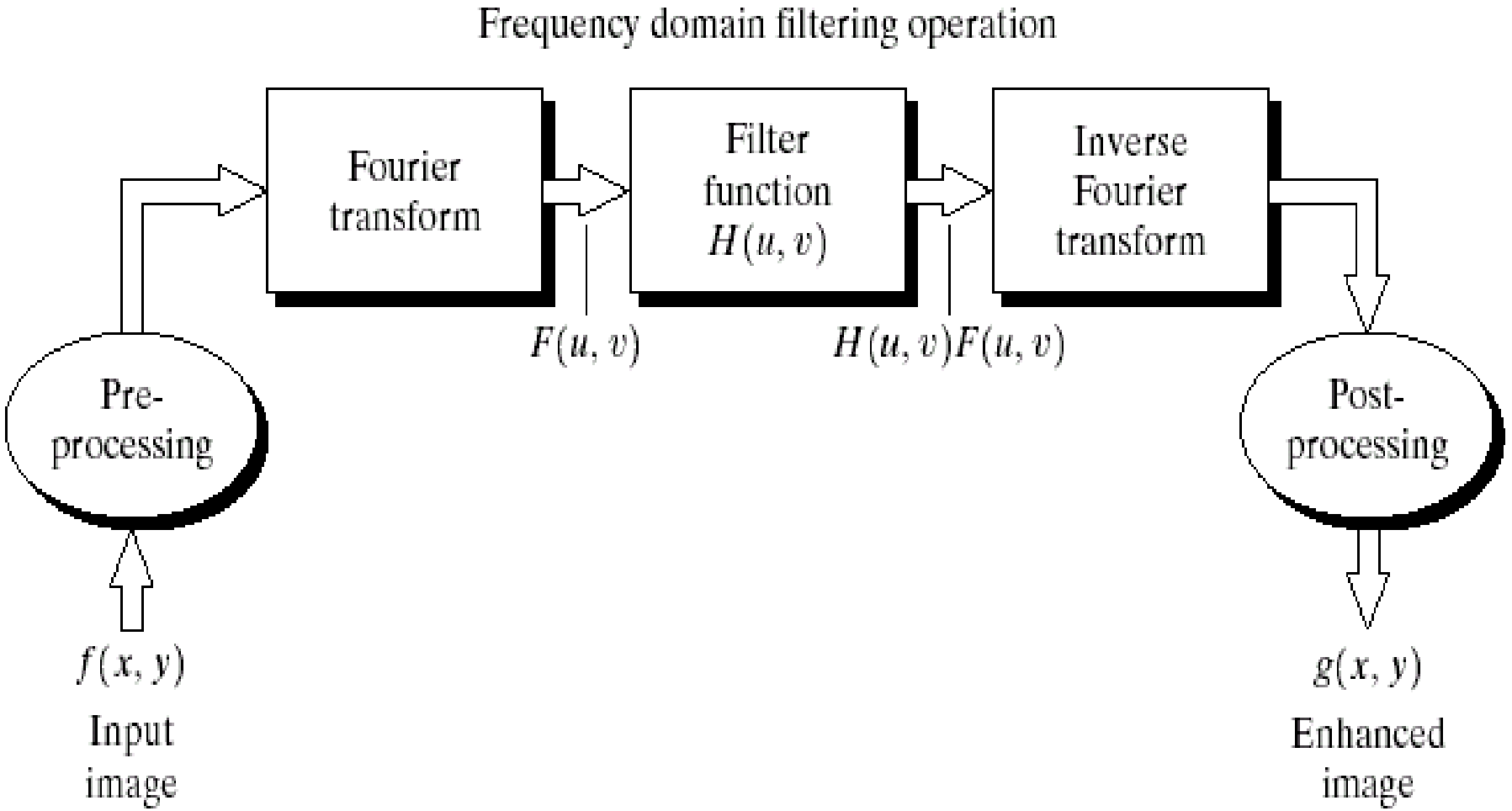


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Low Pass Filter (LPF) dalam domain frekuensi

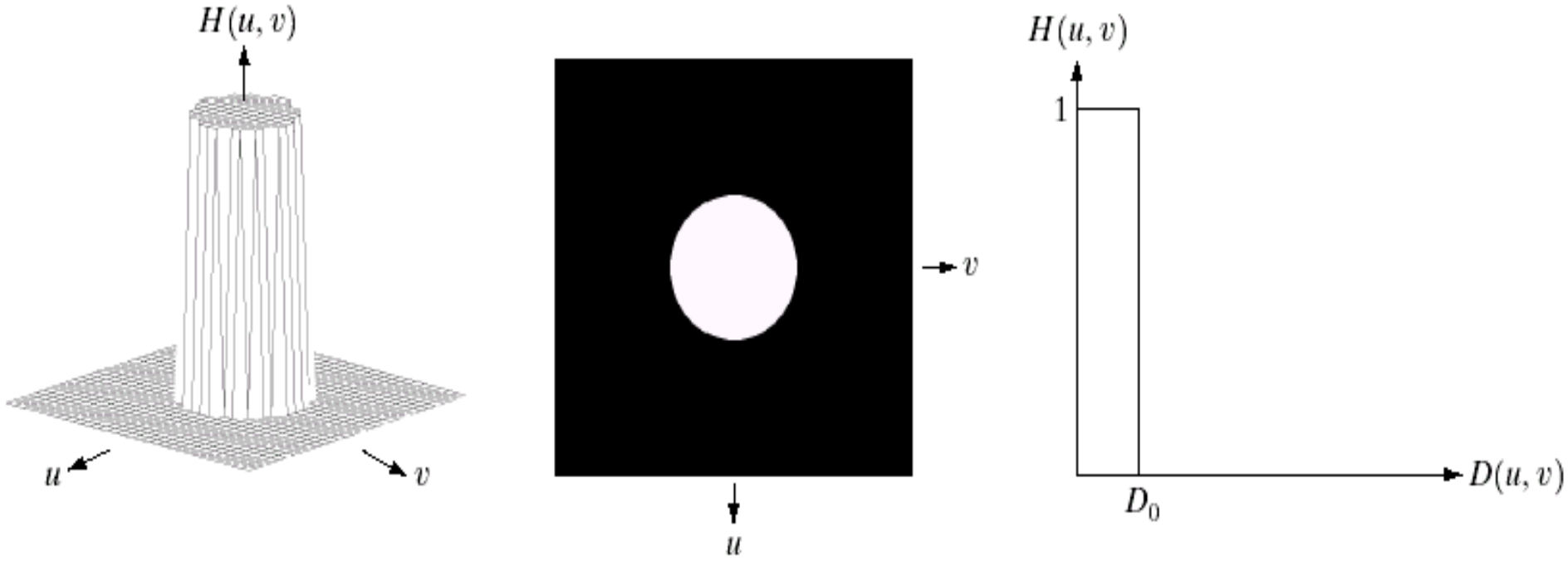
TABLE 4.4

Lowpass filters. D_0 is the cutoff frequency and n is the order of the Butterworth filter.

Ideal	Butterworth	Gaussian
$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$	$H(u,v) = \dfrac{1}{1 + [D(u,v)/D_0]^{2n}}$	$H(u,v) = e^{-D^2(u,v)/2D_0^2}$

D (u,v) adalah jarak antara titik (u,v) dengan titik pusat

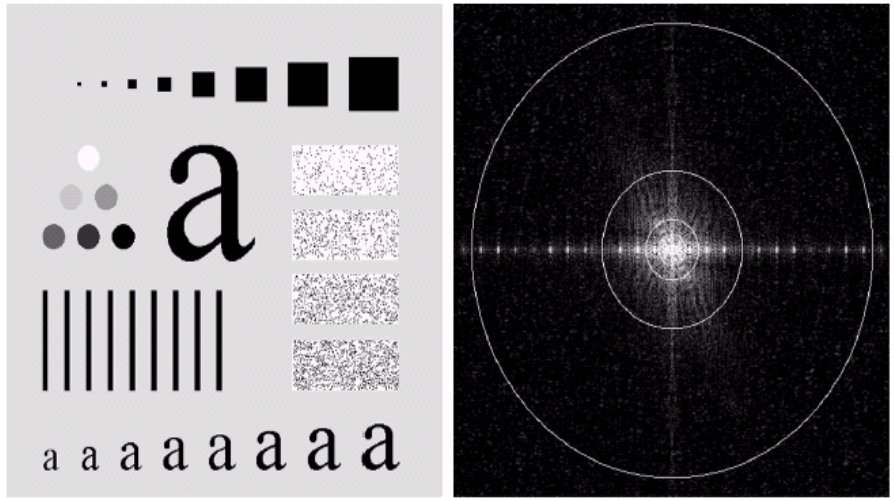
Ideal LPF



a b c

FIGURE 4.10 (a) Perspective plot of an ideal lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

Contoh ILPF



a b

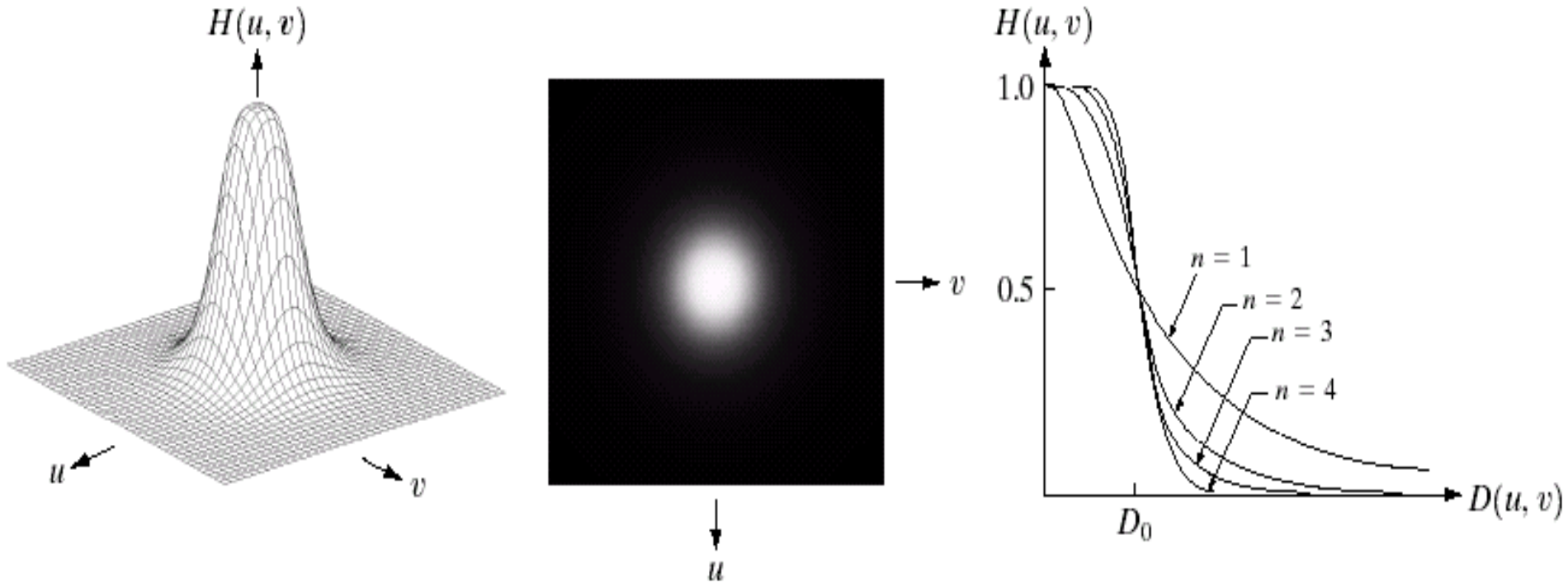
FIGURE 4.11 (a) An image of size 500 × 500 pixels and (b) its Fourier spectrum. The superimposed circles have radii values of 5, 15, 30, 80, and 230, which enclose 92.0, 94.6, 96.4, 98.0, and 99.5% of the image power, respectively.



a b
c d
e f

FIGURE 4.12 (a) Original image. (b)–(f) Results of ideal lowpass filtering with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). The power removed by these filters was 8, 5.4, 3.6, 2, and 0.5% of the total, respectively.

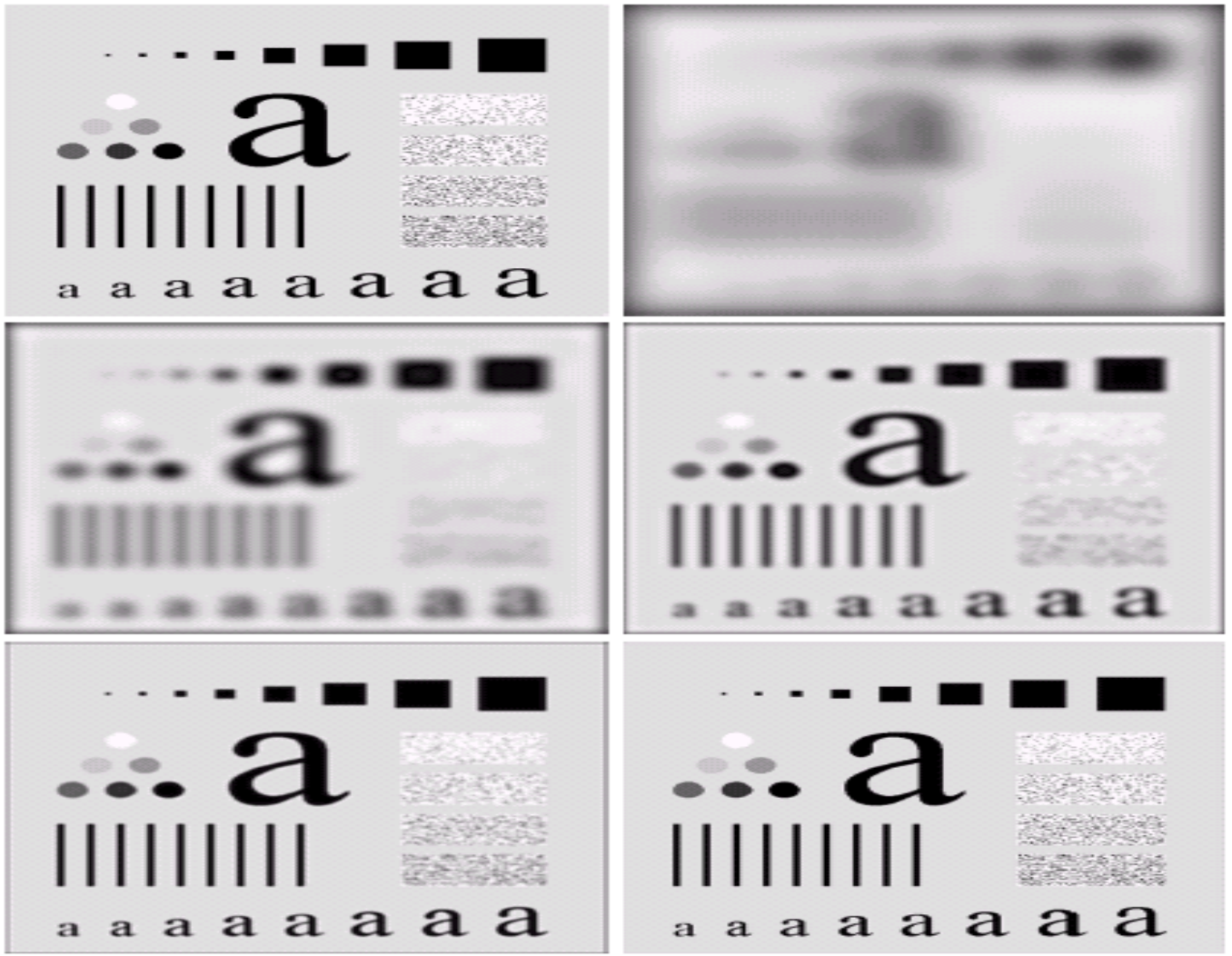
Butterworth LPF



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.

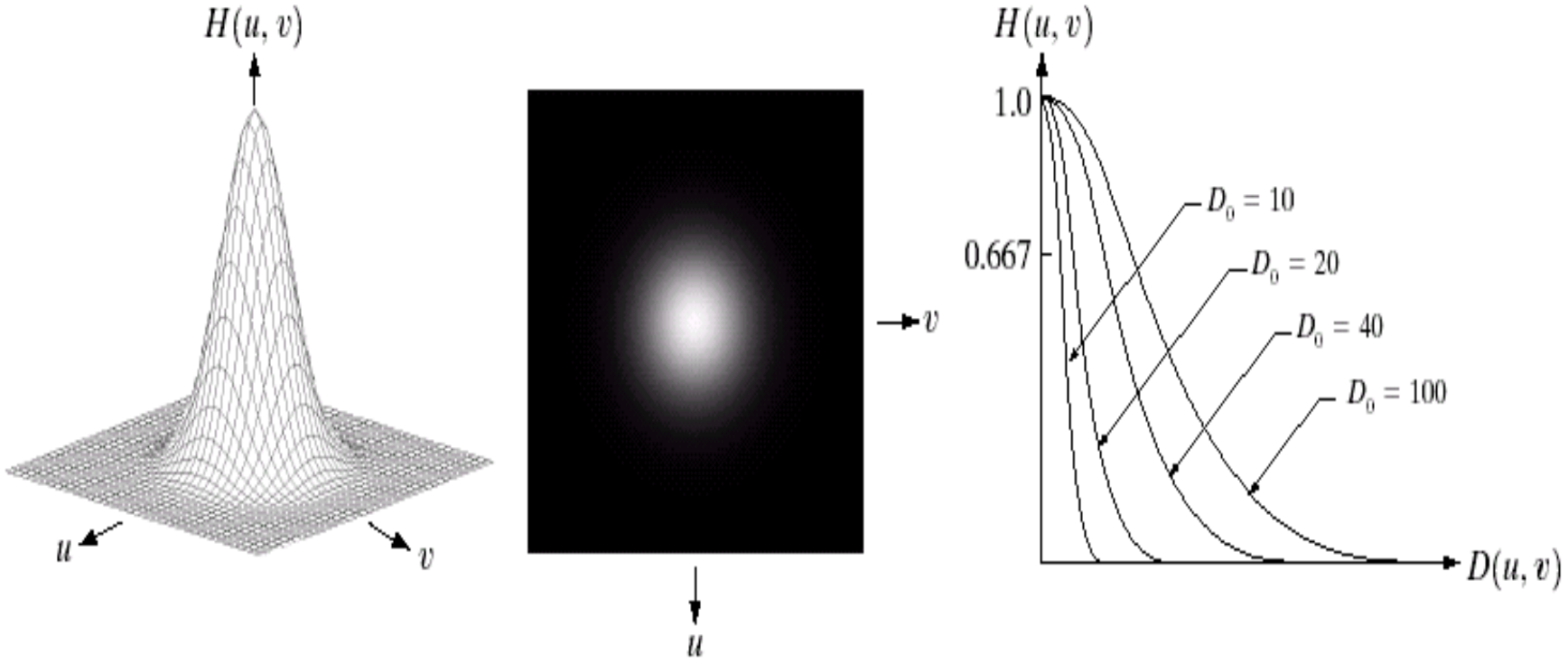
Contoh BLPF



a b
c d
e f

FIGURE 4.15 (a) Original image. (b)–(f) Results of filtering with BLPFs of order 2, with cutoff frequencies at radii of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Fig. 4.12.

Gaussian LPF (GLPF)



a b c

FIGURE 4.17 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.

Contoh GLPF

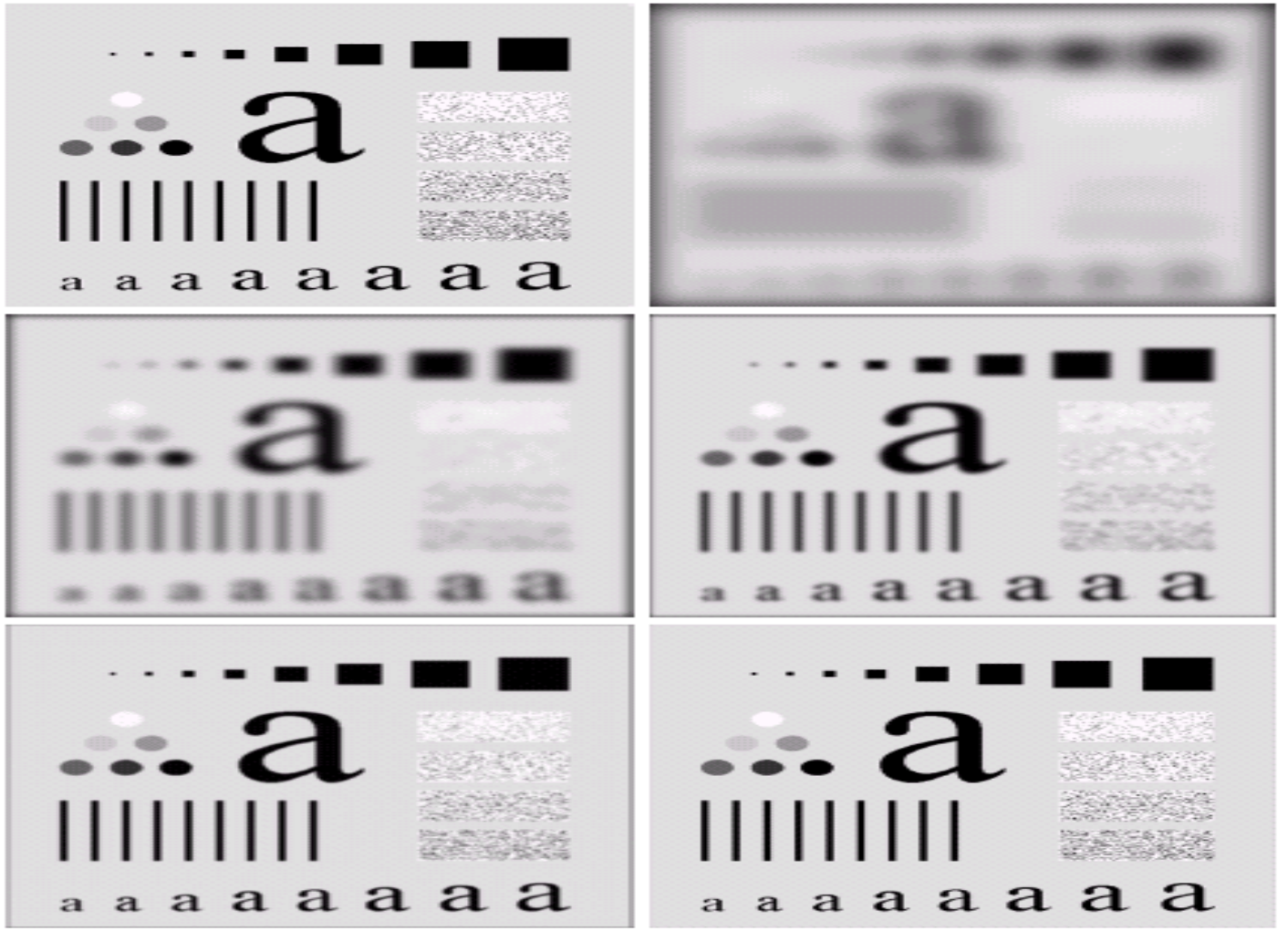


FIGURE 4.18 (a) Original image. (b)–(f) Results of filtering with Gaussian lowpass filters with cutoff frequencies set at radii values of 5, 15, 30, 80, and 230, as shown in Fig. 4.11(b). Compare with Figs. 4.12 and 4.15.

a b
c d
e f

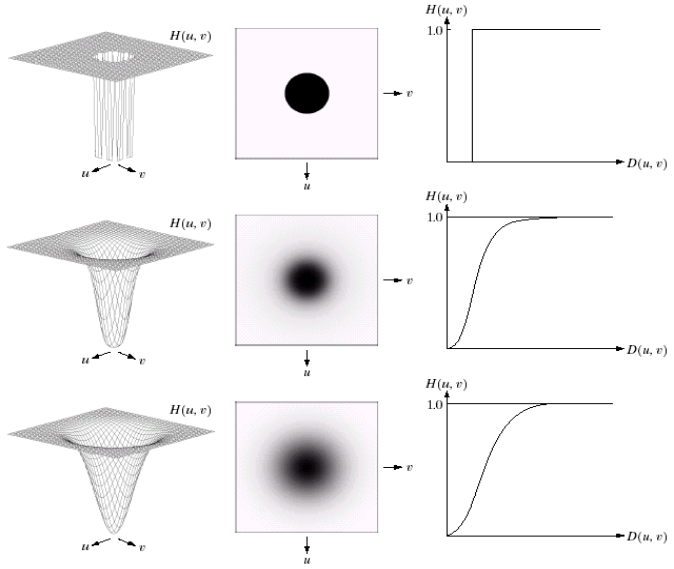
HPF dalam domain Frekuensi

Ideal highpass filter

$$H(u,v) = \begin{cases} 0 \text{ if } D(u,v) \le D_0 \\ 1 \text{ if } D(u,v) > D_0 \end{cases}$$

Butterworth highpass filter

$$H(u,v) = \frac{1}{1+\left[D_0 / D(u,v)\right]^{2n}}$$



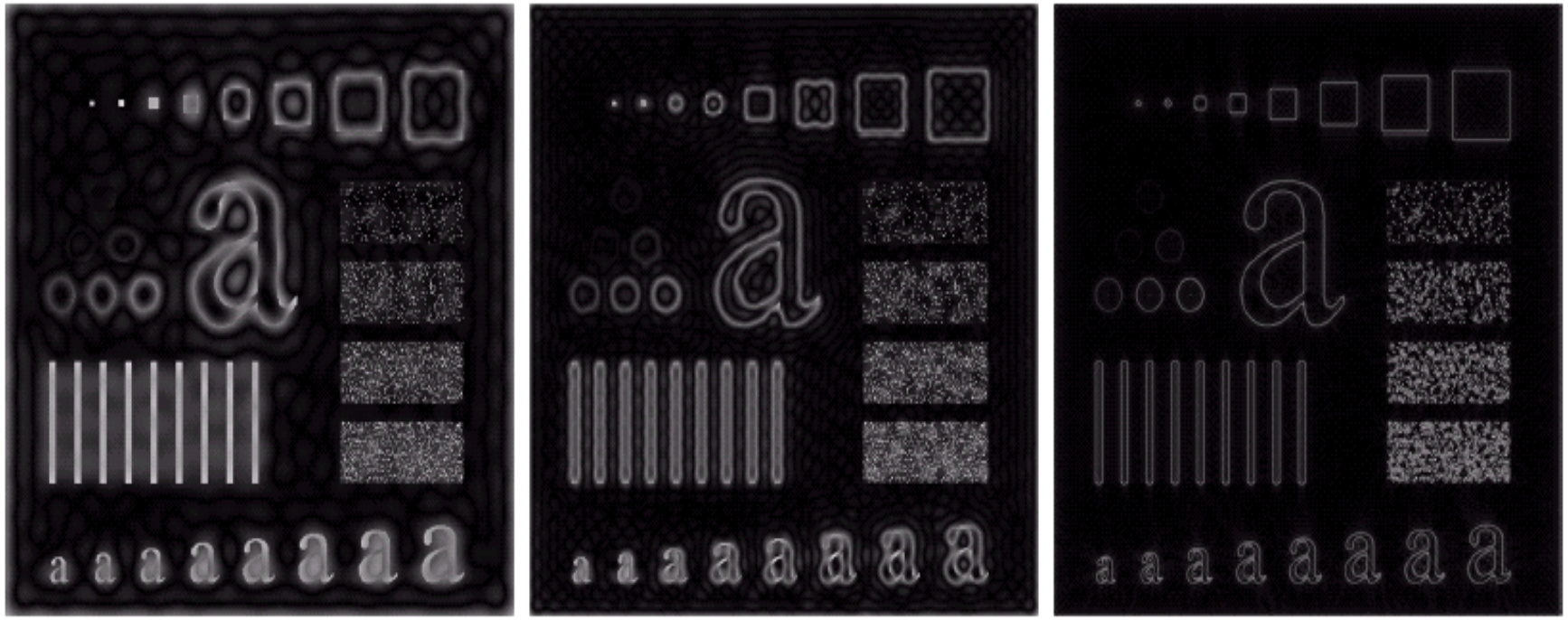
a b c
d e f
g h i

FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Gaussian highpass filter

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

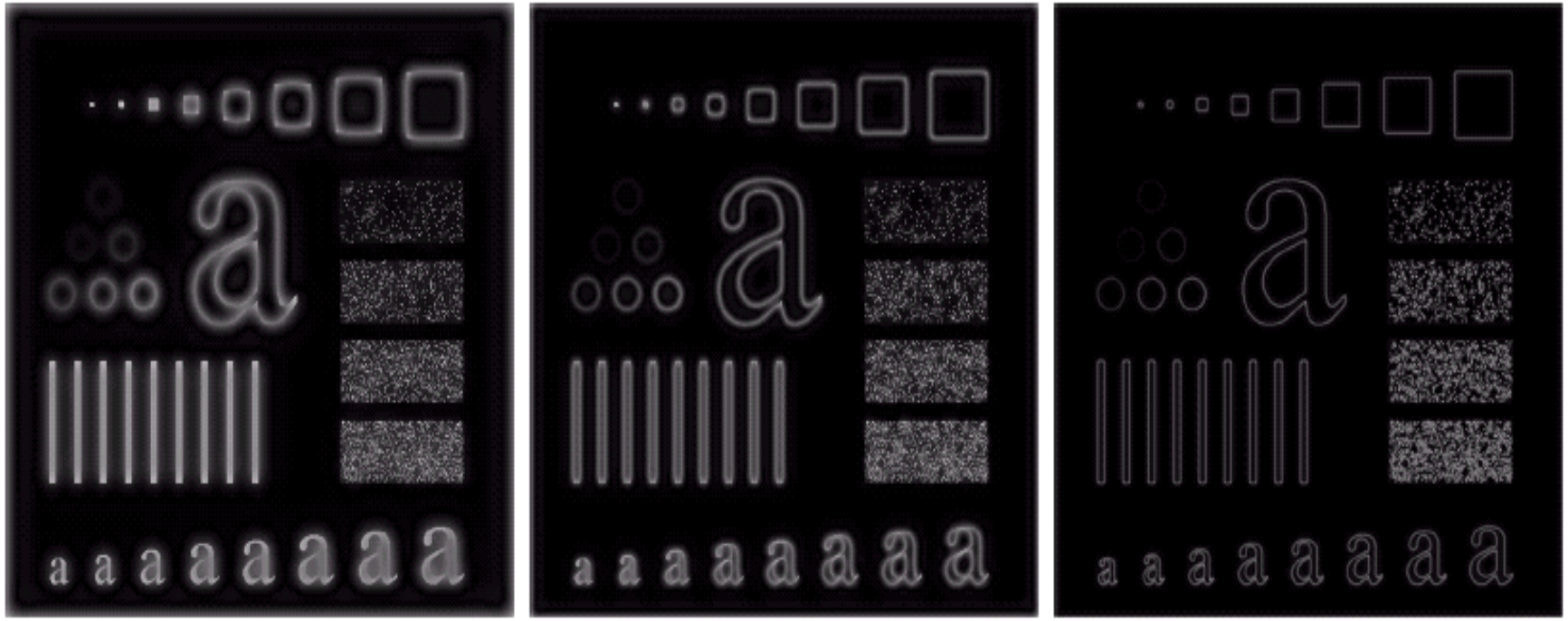
Contoh IHPF



a b c

FIGURE 4.24 Results of ideal highpass filtering the image in Fig. 4.11(a) with $D_0 = 15$, 30, and 80, respectively. Problems with ringing are quite evident in (a) and (b).

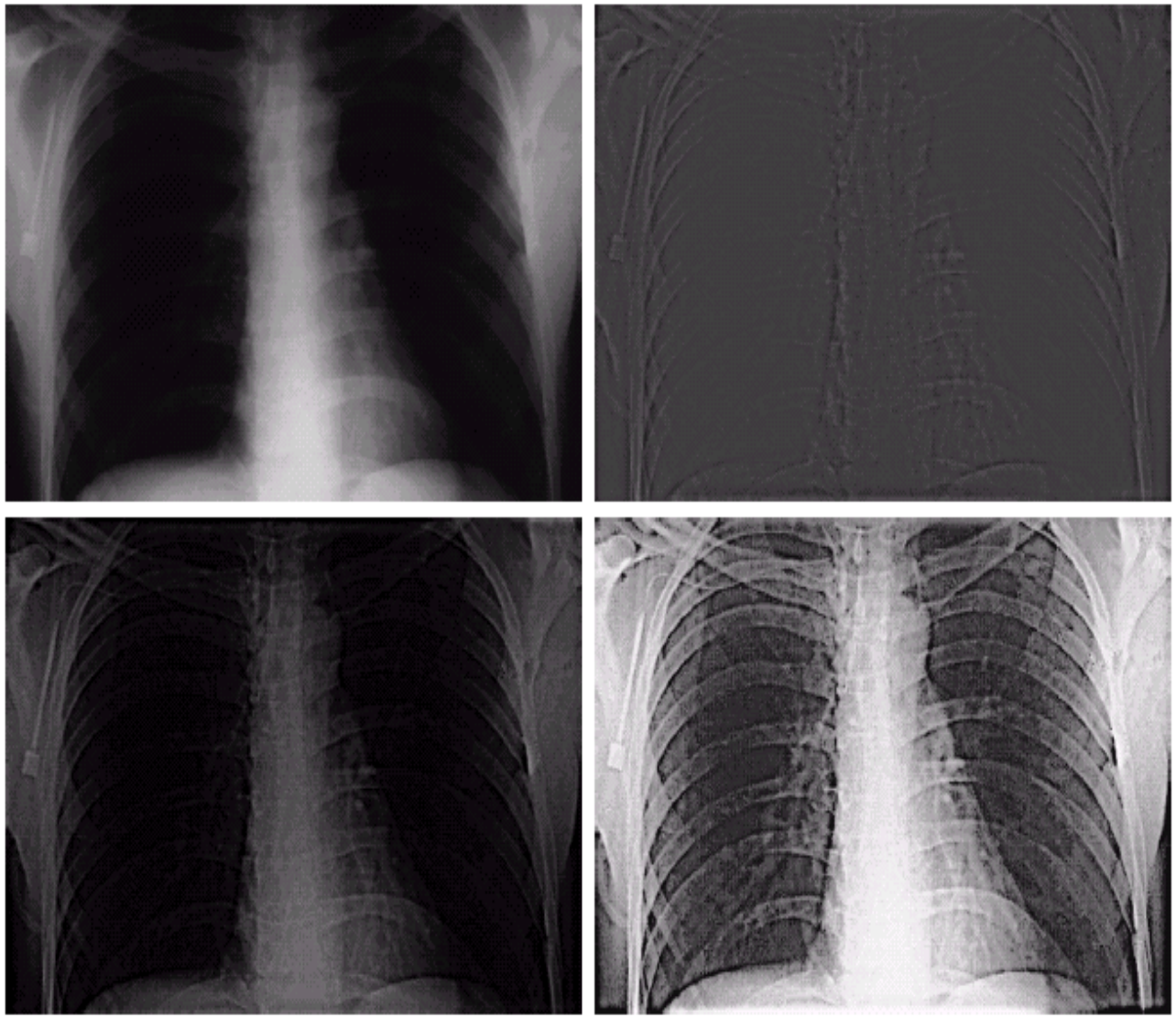
Contoh BHPF



a b c

FIGURE 4.25 Results of highpass filtering the image in Fig. 4.11(a) using a BHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. These results are much smoother than those obtained with an ILPF.

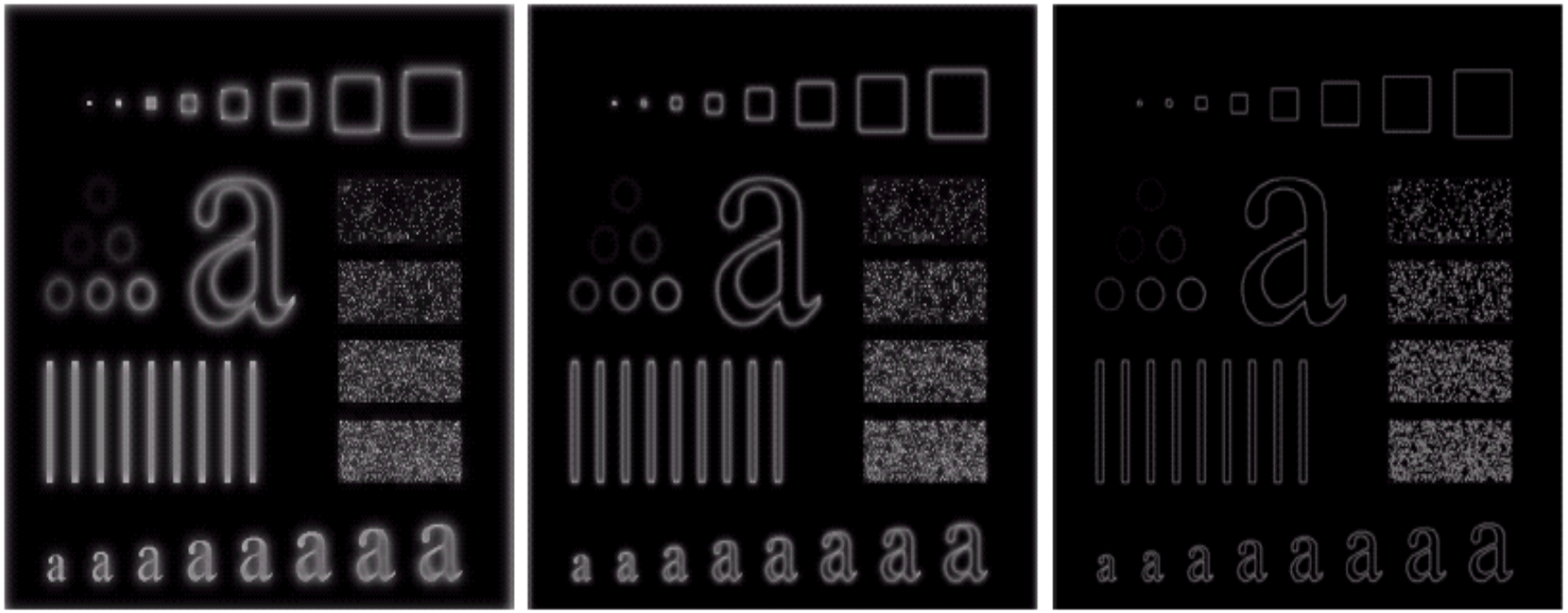
Contoh



a b
c d

FIGURE 4.30
(a) A chest X-ray image. (b) Result of Butterworth highpass filtering. (c) Result of high-frequency emphasis filtering. (d) Result of performing histogram equalization on (c). (Original image courtesy Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School.)

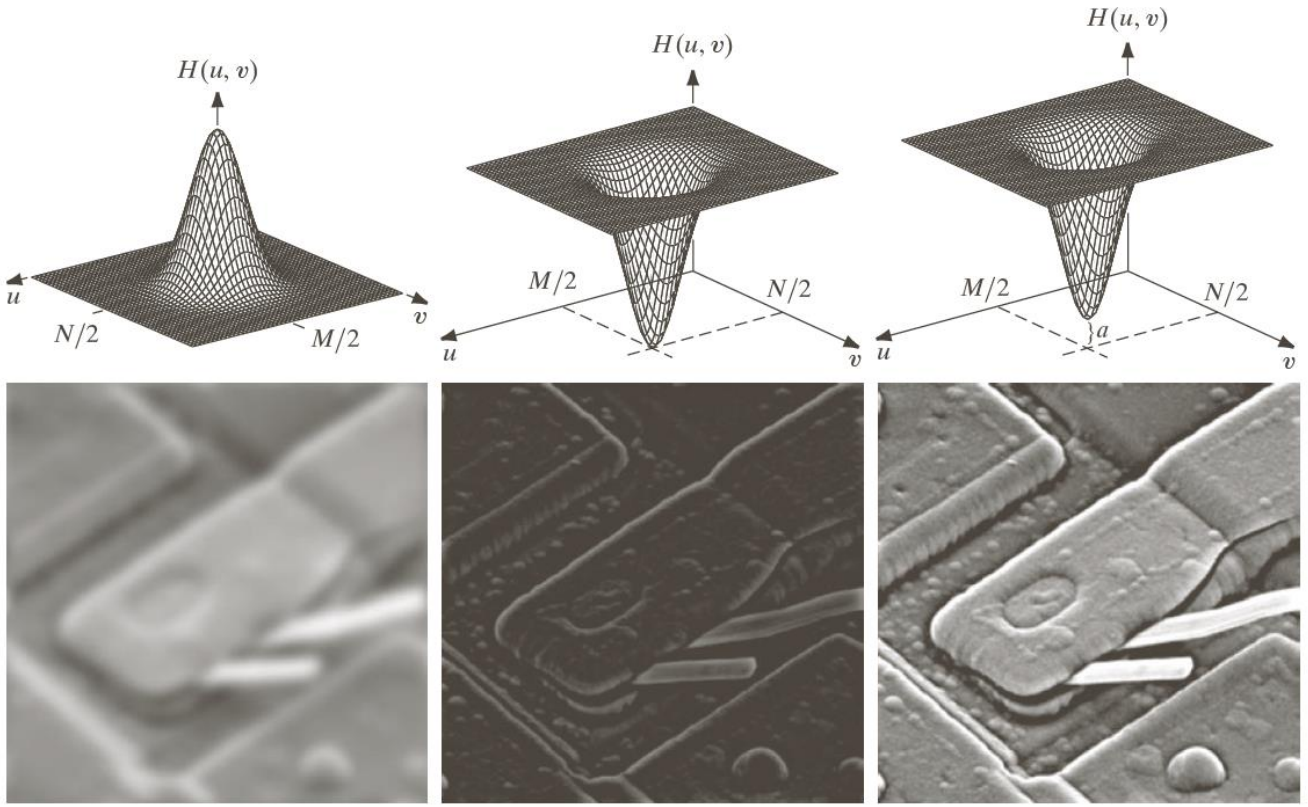
Contoh GHPF



a b c

FIGURE 4.26 Results of highpass filtering the image of Fig. 4.11(a) using a GHPF of order 2 with $D_0 = 15$, 30, and 80, respectively. Compare with Figs. 4.24 and 4.25.

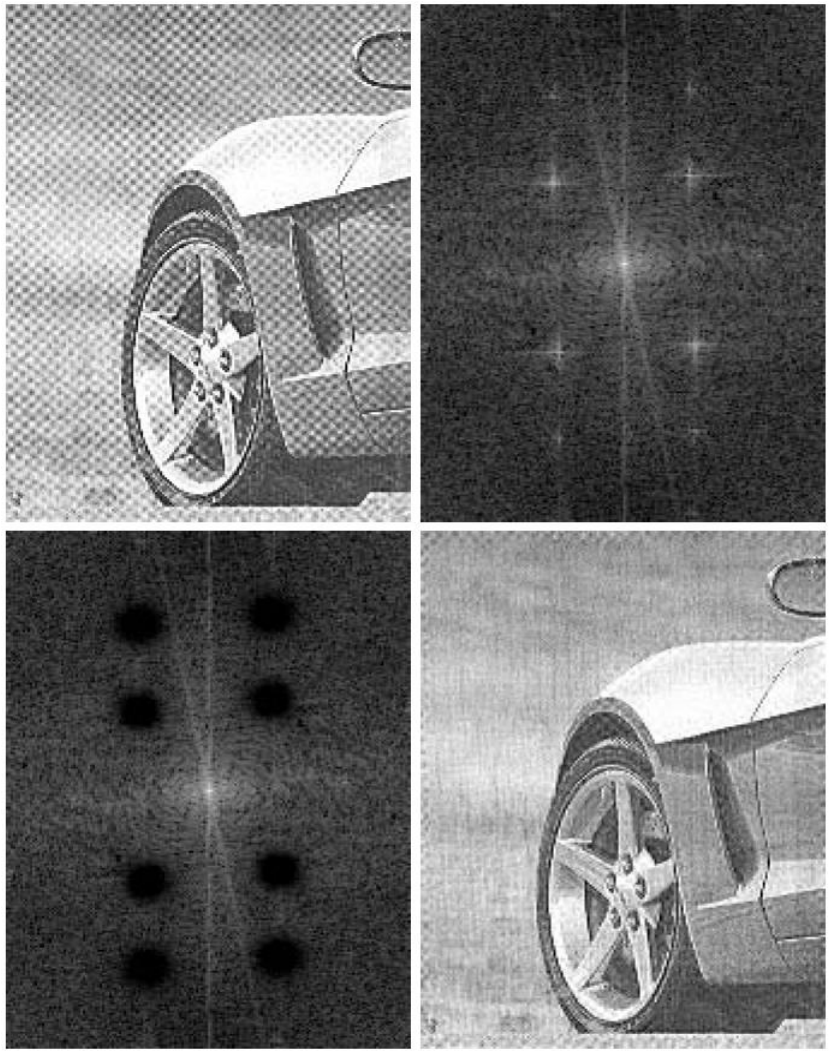
FILTER DAN CITRA HASIL



a b c
d e f

FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).

Hasil Butterworth Notch Reject Filter



a b
c d

FIGURE 4.64
(a) Sampled newspaper image showing a moiré pattern.
(b) Spectrum.
(c) Butterworth notch reject filter multiplied by the Fourier transform.
(d) Filtered image.

Homomorphic Filter

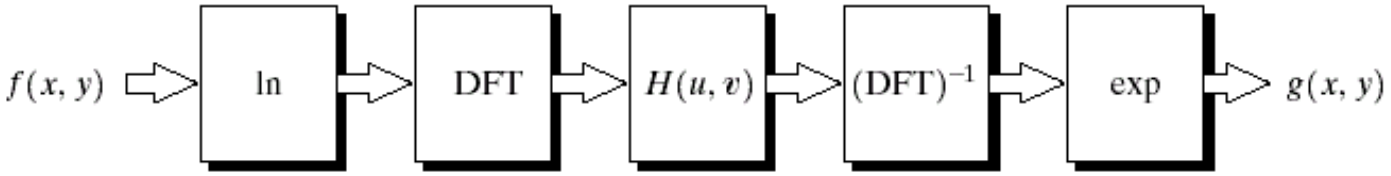


FIGURE 4.31 Homomorphic filtering approach for image enhancement.

$$H(u,v) = (\gamma_H - \gamma_L)\left[1 - e^{-cD^2(u,v)/D_0^2}\right] + \gamma_L$$

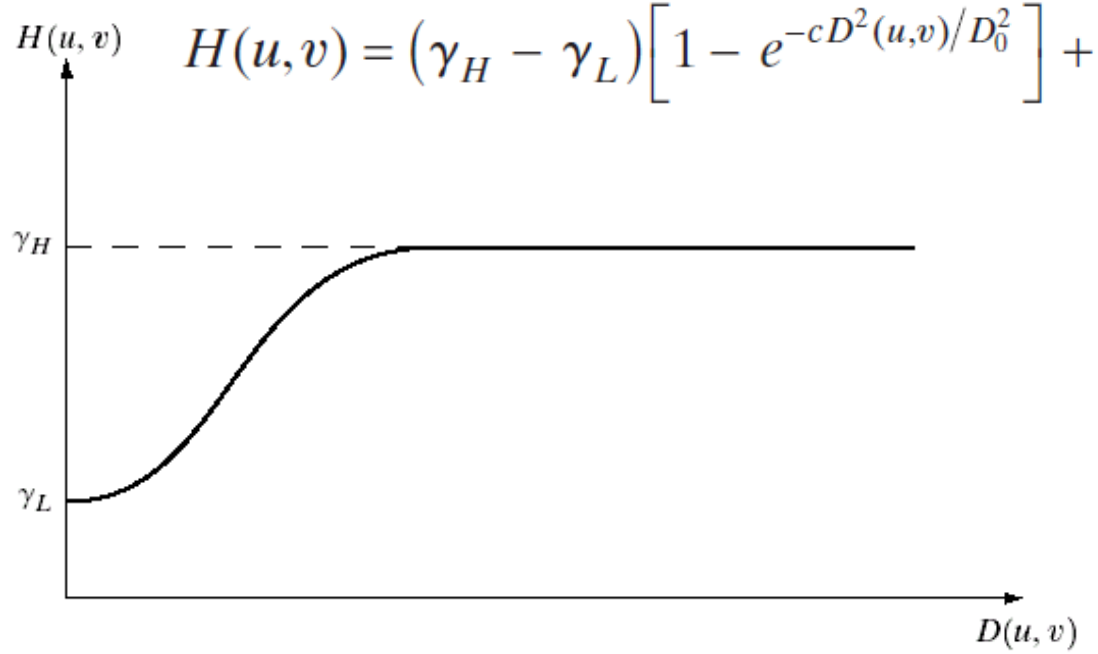
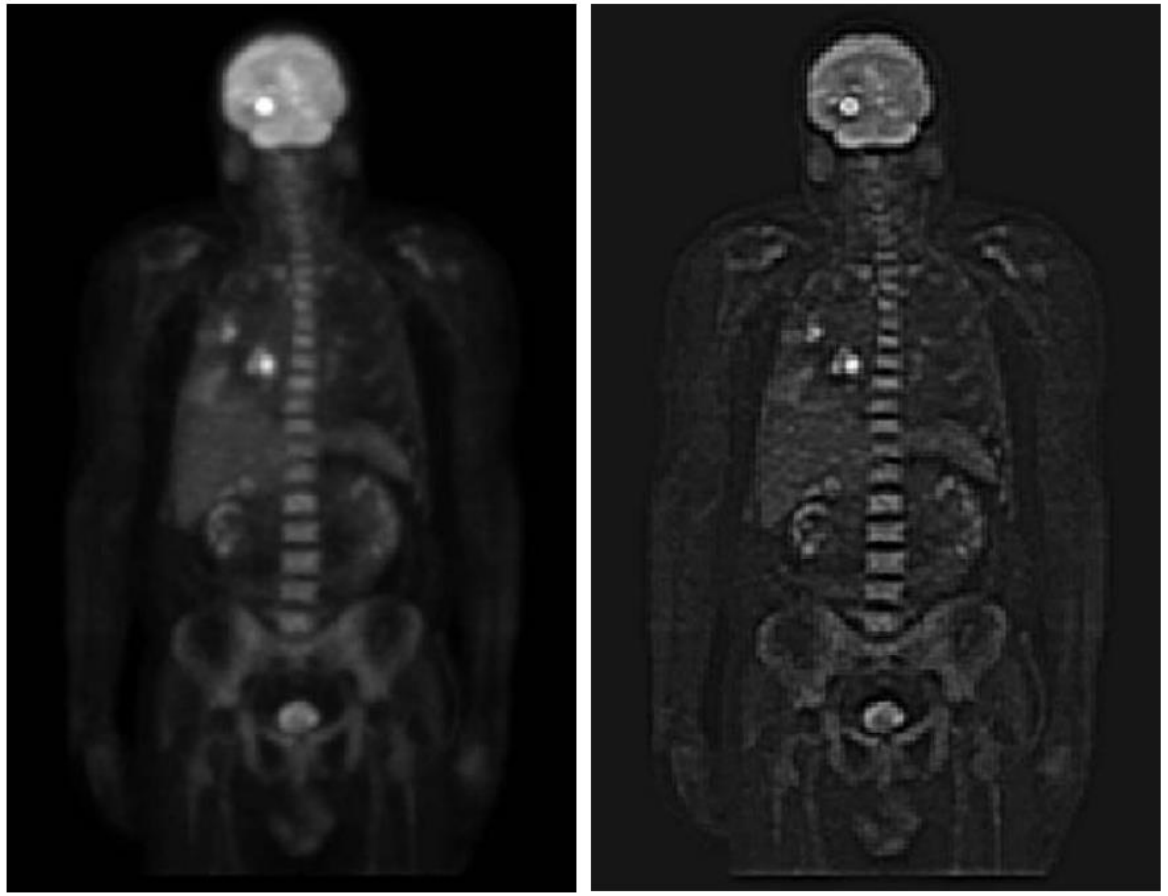


FIGURE 4.32 Cross section of a circularly symmetric filter function. $D(u, v)$ is the distance from the origin of the centered transform.

Hasil Homomorphic Filter



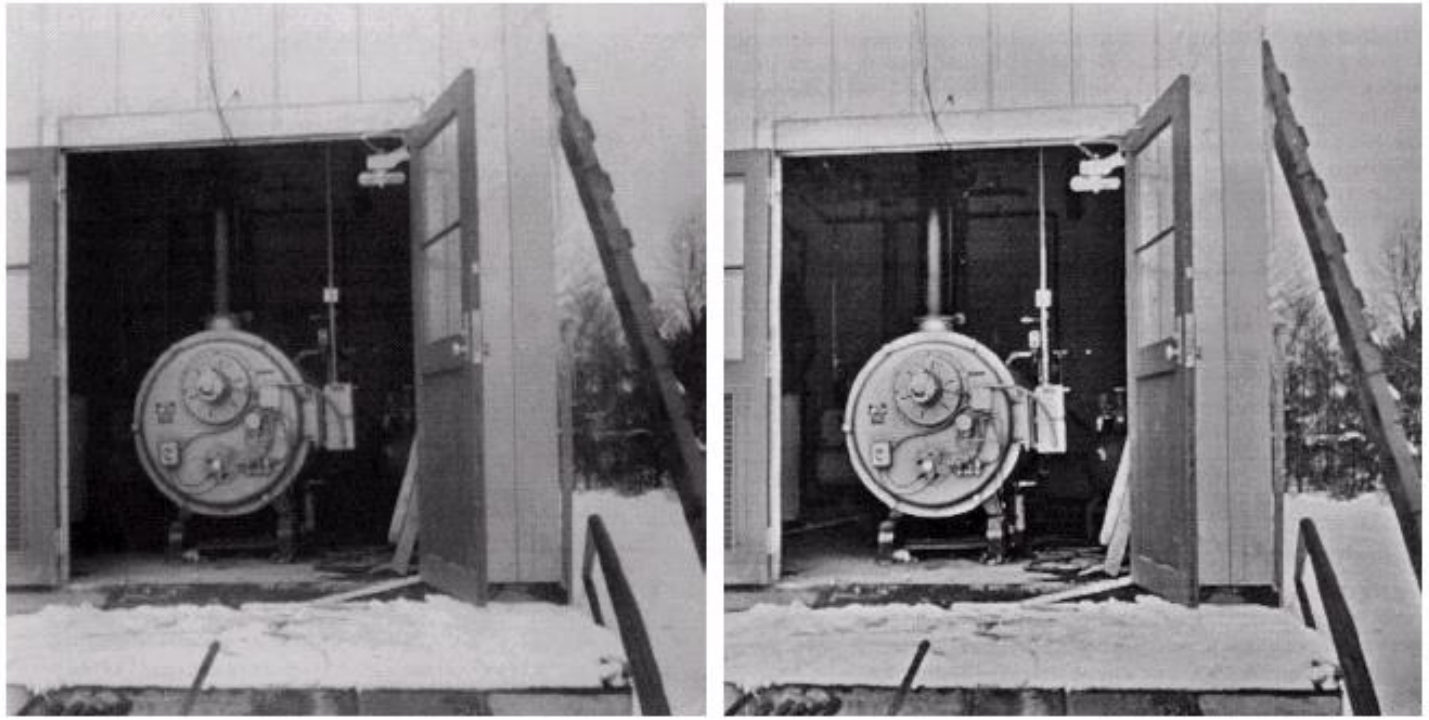
a b

FIGURE 4.62
(a) Full body PET scan. (b) Image enhanced using homomorphic filtering. (Original image courtesy of Dr. Michael E. Casey, CTI PET Systems.)

Hasil Homomorphic filter

a b

FIGURE 4.33
(a) Original image. (b) Image processed by homomorphic filtering (note details inside shelter). (Stockham.)




THANK YOU